

YFG FENDER HOMOLOGATION & BOP CHECKING MACHINE

CV-X300 Series

STD FENDER BOP CHECKING



MMLI

Index

1. Display Type and View	8
1.1. Operation Screen	8
1.1.1. Display View and Operation	8
1.1.2. Operation Screen List.....	11
1.2. Setting Screen	13
1.2.1. Display View and Operation	13
2. Camera Adjustment	16
2.1. (CAM1) Camera Settings.....	16
2.1.1. Defaults List of Camera Parameters	16
2.2. (CAM2) Camera Settings.....	16
2.2.1. Defaults List of Camera Parameters	16
3. Tool Adjustment.....	18
3.1. Tool List	18
3.2. (CAM1) [T101]NITTO CAP	19
3.2.1. Summary	19
3.2.2. Defaults List of Parameters	19
3.2.3. Tool Edit Screen View	20
3.2.4. Setting the Inspection Region.....	21
3.2.5. Adjustment for Detection Conditions	22
3.2.6. Adjustment for Judgment Conditions	23
3.3. (CAM1) [T102]GASKET-1.....	24
3.3.1. Summary	24
3.3.2. Defaults List of Parameters	24

3.3.3. Tool Edit Screen View	25
3.3.4. Setting the Inspection Region.....	26
3.3.5. Adjustment for Detection Conditions	27
3.3.6. Adjustment for Judgment Conditions	28
3.4. (CAM1) [T103]GASKET-2.....	29
3.4.1. Summary	29
3.4.2. Defaults List of Parameters	29
3.4.3. Tool Edit Screen View	30
3.4.4. Setting the Inspection Region.....	31
3.4.5. Adjustment for Detection Conditions	32
3.4.6. Adjustment for Judgment Conditions	33
3.5. (CAM1) [T104]SQUARE GASKET.....	34
3.5.1. Summary	34
3.5.2. Defaults List of Parameters	34
3.5.3. Tool Edit Screen View	35
3.5.4. Setting the Inspection Region.....	36
3.5.5. Adjustment for Detection Conditions	37
3.5.6. Adjustment for Judgment Conditions	38
3.6. (CAM1) [T105]FINISHER	39
3.6.1. Summary	39
3.6.2. Defaults List of Parameters	39
3.6.3. Tool Edit Screen View	40
3.6.4. Setting the Inspection Region.....	41
3.6.5. Adjustment for Binary.....	42

3.6.6. Adjustment for Detection Conditions	43
3.6.7. Adjustment for Judgment Conditions	44
3.7. (CAM1) [T106]SCREW-1.....	45
3.7.1. Summary	45
3.7.2. Defaults List of Parameters	45
3.7.3. Tool Edit Screen View	46
3.7.4. Setting the Inspection Region.....	47
3.7.5. Adjustment for Binary.....	48
3.7.6. Adjustment for Detection Conditions	49
3.7.7. Adjustment for Judgment Conditions	50
3.8. (CAM1) [T107]SCREW-2.....	51
3.8.1. Summary	51
3.8.2. Defaults List of Parameters	51
3.8.3. Tool Edit Screen View	52
3.8.4. Setting the Inspection Region.....	53
3.8.5. Adjustment for Binary.....	54
3.8.6. Adjustment for Detection Conditions	55
3.8.7. Adjustment for Judgment Conditions	56
3.9. (CAM2) [T201]NITTO CAP	57
3.9.1. Summary	57
3.9.2. Defaults List of Parameters	57
3.9.3. Tool Edit Screen View	58
3.9.4. Setting the Inspection Region.....	59
3.9.5. Adjustment for Detection Conditions	60

3.9.6. Adjustment for Judgment Conditions	61
3.10. (CAM2) [T202]GASKET-1.....	62
3.10.1. Summary	62
3.10.2. Defaults List of Parameters	62
3.10.3. Tool Edit Screen View	63
3.10.4. Setting the Inspection Region.....	64
3.10.5. Adjustment for Detection Conditions	65
3.10.6. Adjustment for Judgment Conditions	66
3.11. (CAM2) [T203]GASKET-2.....	67
3.11.1. Summary	67
3.11.2. Defaults List of Parameters	67
3.11.3. Tool Edit Screen View	68
3.11.4. Setting the Inspection Region.....	69
3.11.5. Adjustment for Detection Conditions	70
3.11.6. Adjustment for Judgment Conditions	71
3.12. (CAM2) [T204]SQUARE GASKET	72
3.12.1. Summary	72
3.12.2. Defaults List of Parameters	72
3.12.3. Tool Edit Screen View	73
3.12.4. Setting the Inspection Region.....	74
3.12.5. Adjustment for Detection Conditions	75
3.12.6. Adjustment for Judgment Conditions	76
3.13. (CAM2) [T205]FINISHER.....	77
3.13.1. Summary	77

3.13.2. Defaults List of Parameters	77
3.13.3. Tool Edit Screen View	78
3.13.4. Setting the Inspection Region.....	79
3.13.5. Adjustment for Binary	80
3.13.6. Adjustment for Detection Conditions	81
3.13.7. Adjustment for Judgment Conditions	82
3.14. (CAM2) [T206]SCREW-1.....	83
3.14.1. Summary	83
3.14.2. Defaults List of Parameters	83
3.14.3. Tool Edit Screen View	84
3.14.4. Setting the Inspection Region.....	85
3.14.5. Adjustment for Binary	86
3.14.6. Adjustment for Detection Conditions	87
3.14.7. Adjustment for Judgment Conditions	88
3.15. (CAM2) [T207]SCREW-2.....	89
3.15.1. Summary	89
3.15.2. Defaults List of Parameters	89
3.15.3. Tool Edit Screen View	90
3.15.4. Setting the Inspection Region.....	91
3.15.5. Adjustment for Binary	92
3.15.6. Adjustment for Detection Conditions	93
3.15.7. Adjustment for Judgment Conditions	94
4. Supplement.....	95
4.1. Extract Colors Settings	95

4.2. Image Enhance Settings.....	97
5. Appendix	98
5.1. Program Export/Import	98
5.1.1. Preparation	98
5.1.2. Archived Image	100
5.1.3. Export.....	101
5.1.4. Import	107
5.1.5. Exchange of Setting File	110
5.2. Outputting Total Status	111
5.2.1. Performing Wiring	111
5.2.2. Confirming External Terminal Connection Status (I/O Monitor)	113
5.2.3. Setting the Tools to Include in Total Status Judgment.....	114
5.2.4. Setting OR Terminal Output Mode.....	115
5.2.5. Timing Chart.....	116

1. Display Type and View

Describes how to view the display of this controller.

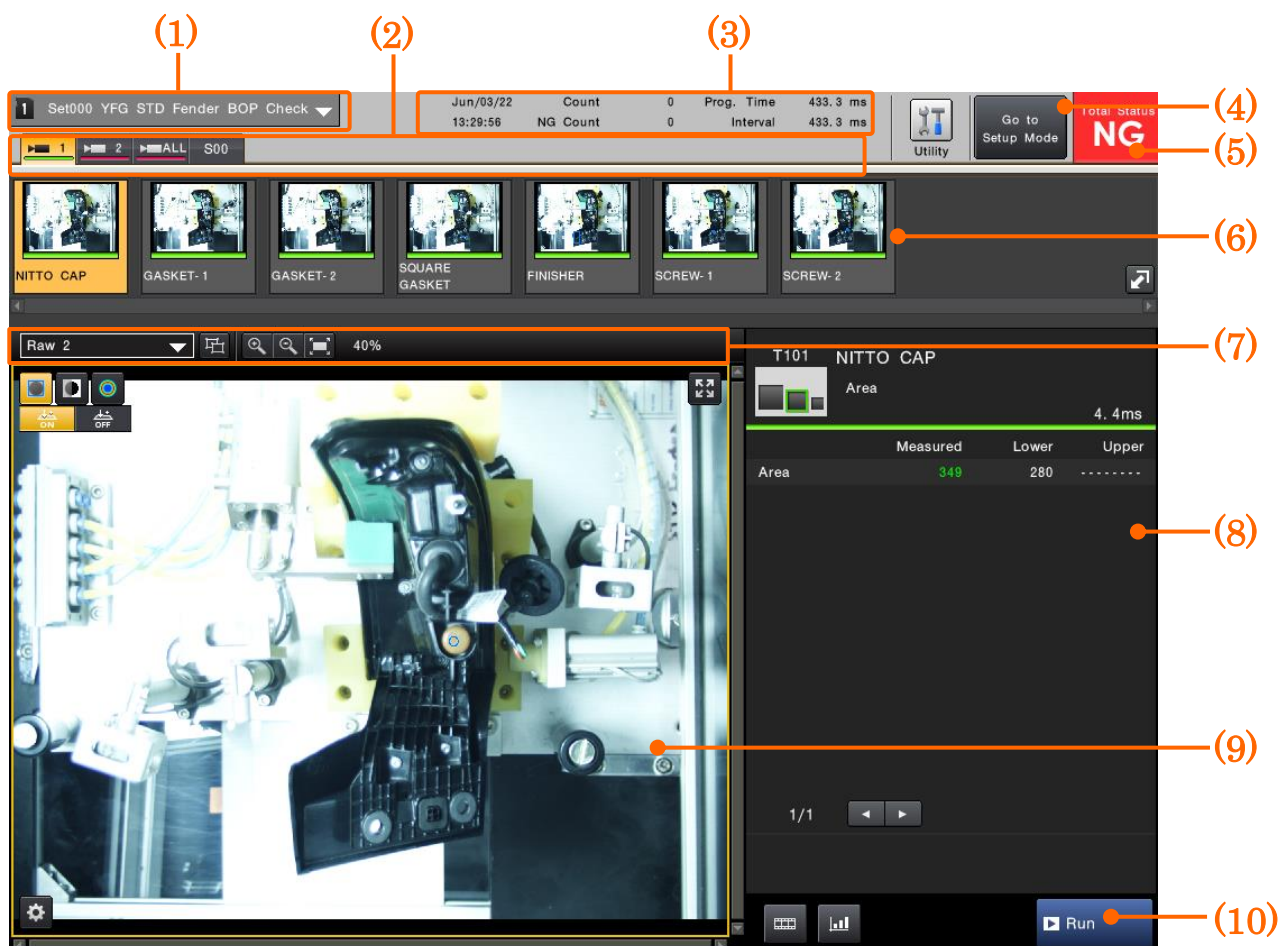
The display varies according to the difference between run and setup modes as well as the operation condition.

1.1. Operation Screen

The display shown in run mode.

1.1.1. Display View and Operation

Items labelled with **Operation Target** come with an operation guide **Operation**. Refer to this guide.



(1)Setting No./Name Display

Displays the setting No. and name currently in operation.

(2)Display Tab Operation Target



These are the tabs used to switch between displays. There are three tabs: CAM tab, ALL tab and Custom tab.

Operation

Clicking the displayed tab switches the operation screen display. Refer to [1.1.2 Operation Screen List](#) for the list of screens that can be switched to in this setting.

(3)Information Display Area

Displays Date, Total Count, Total Status NG Count, Program Time and Interval.

(4)Mode Change Button (Run Mode/Setup Mode) Operation Target



Operation

Pressing the button can switch the mode between the run and setup modes.

(5)Total Status Display



(6)Toolbar (Tool List) Operation Target

Displays the list of the set tools in thumbnail format.

The toolbar thumbnails the tools used in the relevant camera for the CAM tab. For the ALL tab, all of the tools used are displayed.

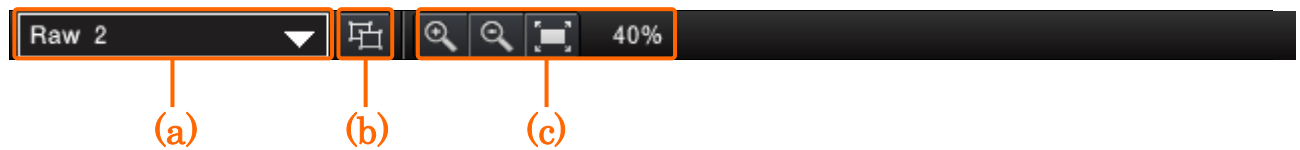
Operation

Click a thumbnail to change the region and measured value displayed in the camera screen to those of the clicked tool.

(7)View Bar Operation Target

Zooms in/out the image displayed or changes the display type.

Operation



(a) Change Display Type

(b) Multiple Regions Display Icon



... Multiple Regions Display OFF



... Multiple Regions Display ON

(c) Display Magnification Icon



... Zoom in



... Zoom out



... Fit



... Current

(8)Measured Value display

Displays the measured value of the selected tool.

(9)Image Display

Displays the image input from the camera.

(10)Run (trigger input) button Operation target

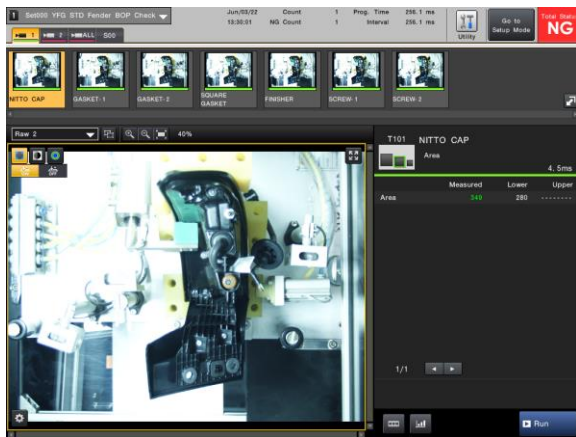
Operation

Press the button to run the measurement.

1.1.2. Operation Screen List

The following are the types screen which can be displayed during operation:

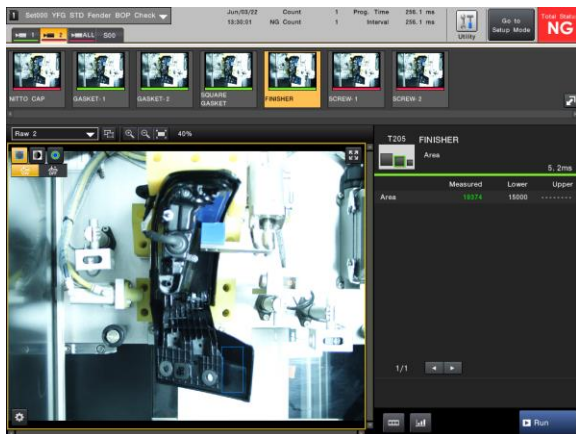
The inspection screen can be switched by clicking on the appropriate CAM tab.



CAM 1 tab

Inspection screen of CAM 1.

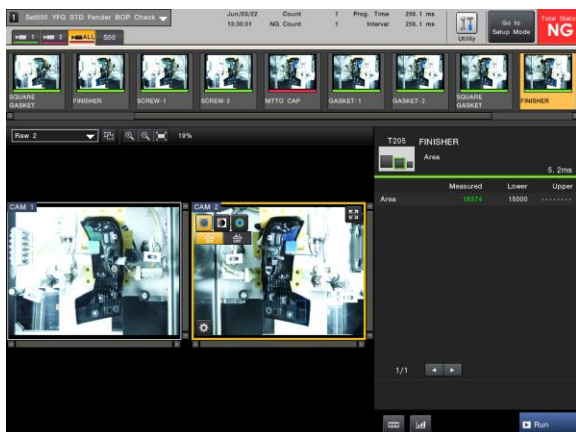
The tools linked with CAM 1 are displayed in the toolbar.



CAM 2 tab

Inspection screen of CAM 2.

The tools linked with CAM 2 are displayed in the toolbar.



CAM ALL Tab

Inspection screen for all cameras.

The toolbar shows all the tools.



S00 Tab

Operation screen S00 [Current Image for 2 Inspections].

The screen layout is as follows.

Display 0 CAM 1 Current Image

Display 1 CAM 2 Current Image

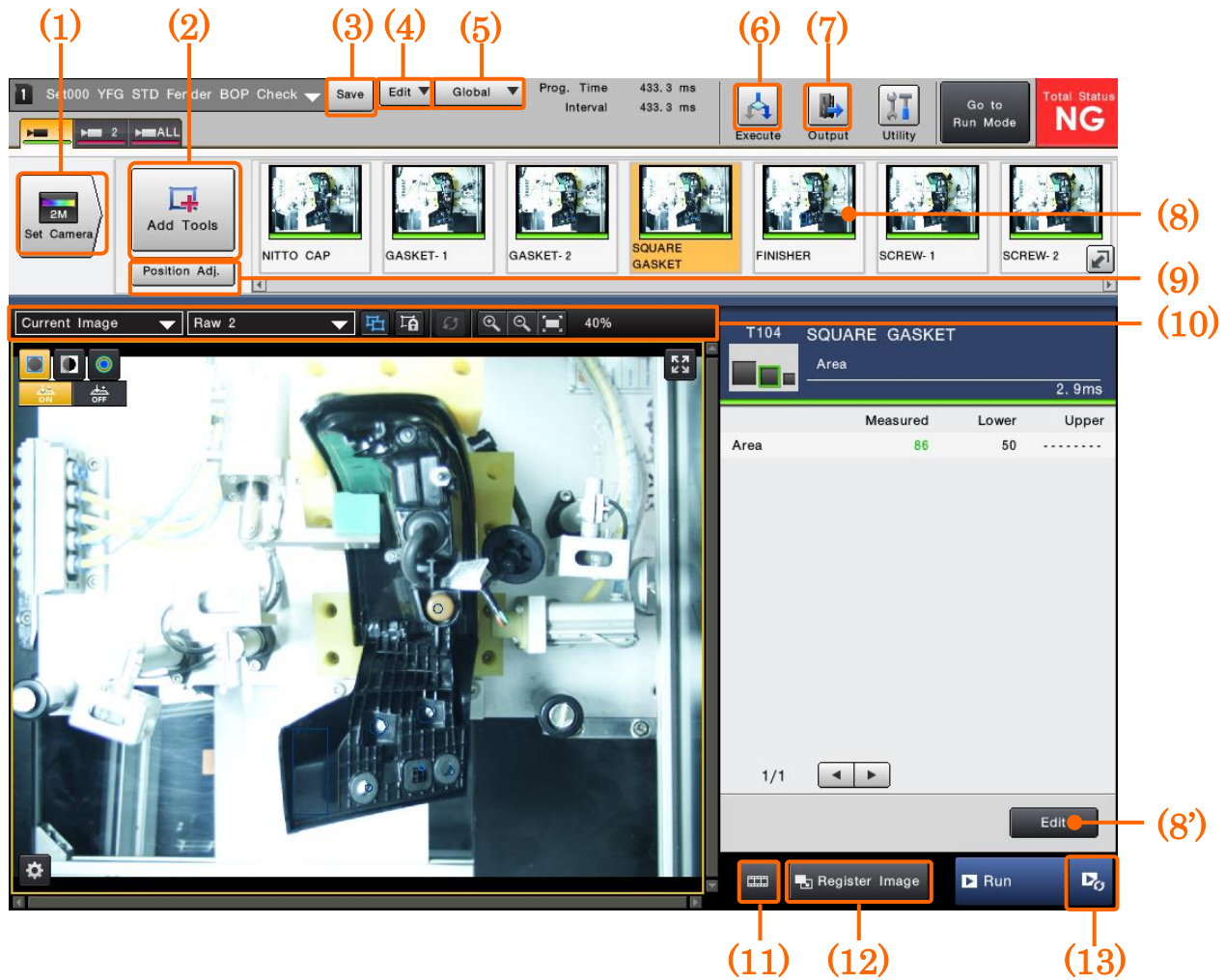
1.2. Setting Screen

The screen shown in setup mode.

1.2.1. Display View and Operation

The item with the **Operation Target** describes the **operation** of the operation guide. Refer to this guide.

Note that the description is omitted for the operation identical to the run mode.



(1)Camera Setting Button Operation Target

Operation

Press the button to show the camera setting screen.

(2)Add Tool Button Operation Target

Operation

Press the button to show the add tool screen.

(3)Save Button Operation Target

Operation

Press the button to save the current program setting.

(4)Edit Button Operation Target

Operation

Press the button to show the menu for editing the program setting.

(5)Global Setting Button Operation Target

Operation

Press the button to show the menu for editing the global setting.

(6)Execute Condition Button Operation Target

Operation

Press the button to show the execution condition setting screen.

(7)Output Setting Button Operation Target

Operation

Press the button to show the output setting screen.

(8)Toolbar (Tool List) Operation Target

Displays the list of the set tools in thumbnail format.

The toolbar thumbnails the tools used in the relevant camera for the CAM tab. For the ALL tab, all the tools used are displayed.

Operation

Select a thumbnail and press the “Edit” button (8'), and the tool specified by the thumbnail can be edited.

Double-clicking the thumbnail performs the same action.

(9)Position Adjustment Button Operation Target

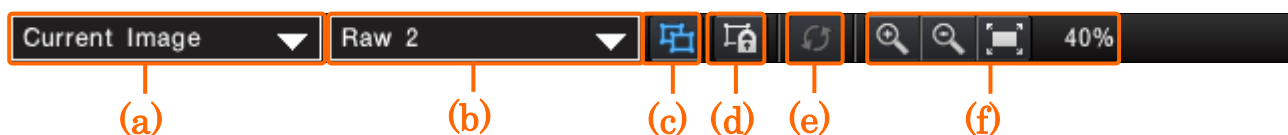
Operation

Press the button to show the position adjustment setting screen.

(10)View Bar Operation Target

Zooms in/out the image displayed or changes the display type.

Operation



(a) Change Display Image

(b) Change Display Type

(c) Multiple Regions Display Icon



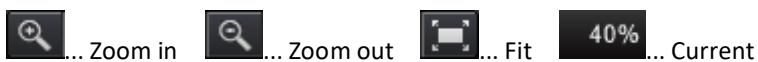
(d) Region Lock Icon



(e) Live Image Icon



(f) Display Magnification Icon



(11)Image Strip Button Operation Target

Operation

Pressing the button can display or hide the image strip.

(12)Image Registration Button Operation Target

Operation

Press the button to show the image registration display.

(13)Repeat Execution Button Operation Target

Operation

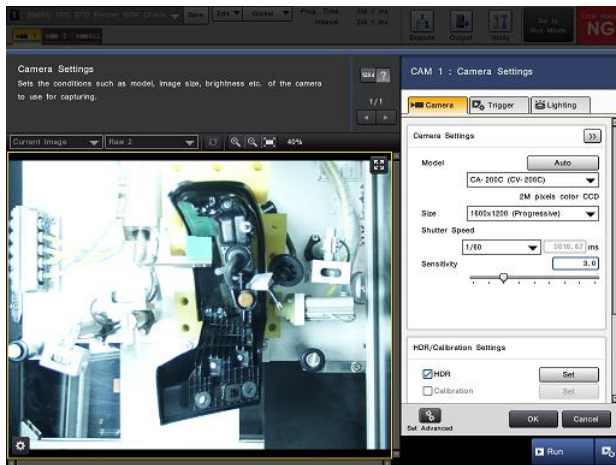
Press the button to execute continuous measurement.

2. Camera Adjustment

2.1. (CAM1) Camera Settings

Sets the conditions such as model, image size, brightness etc. of the camera to use for capturing.

2.1.1. Defaults List of Camera Parameters



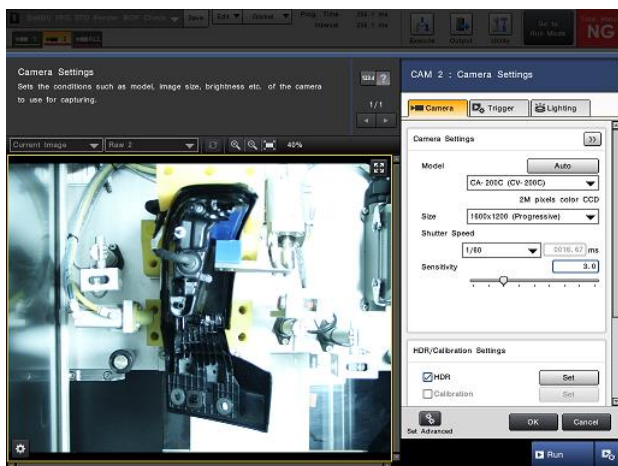
[Camera Settings]

Parameter Name	Defaults	On Change	Memo
Model	CA-200C (CV-200C)		
Size	1600x1200 (Progressive)		
Shutter Speed	1/60 (16.67ms)		
Sensitivity	3		
Mode	Standard Lighting Mode		
HDR	Enabled		

2.2. (CAM2) Camera Settings

Sets the conditions such as model, image size, brightness etc. of the camera to use for capturing.

2.2.1. Defaults List of Camera Parameters



[Camera Settings]

Parameter Name	Defaults	On Change	Memo
Model	CA-200C (CV-200C)		
Size	1600x1200 (Progressive)		
Shutter Speed	1/60 (16.67ms)		
Sensitivity	3		
Mode	Standard Lighting Mode		
HDR	Enabled		

3. Tool Adjustment

By changing to [Setup Mode] using the mode change button, setting adjustment of each tool can be done.

3.1. Tool List

The following tools are used for this setting:

CAM 1			
ID	Tool Name	Memo	Reference
T101	NITTO CAP		p.19
T102	GASKET-1		p.24
T103	GASKET-2		p.29
T104	SQUARE GASKET		p.34
T105	FINISHER		p.39
T106	SCREW-1		p.45
T107	SCREW-2		p.51
CAM 2			
ID	Tool Name	Memo	Reference
T201	NITTO CAP		p.57
T202	GASKET-1		p.62
T203	GASKET-2		p.67
T204	SQUARE GASKET		p.72
T205	FINISHER		p.77
T206	SCREW-1		p.83
T207	SCREW-2		p.89

3.2. (CAM1) [T101]NITTO CAP



3.2.1. Summary

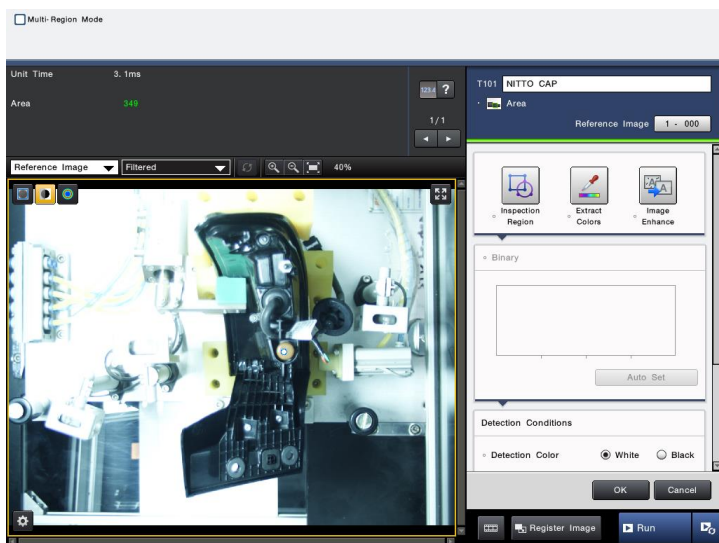
Summary of this tool
Measures the "White" or "Black" area after binarization (black-and-white processing) of the captured image.
Memo

3.2.2. Defaults List of Parameters

Parameter Name	Defaults	On Change	Memo
Detection Color	White		

3.2.3. Tool Edit Screen View

This is the edit screen for the [T101]NITTO CAP tool.



[Inspection Region]

Sets the inspection region. For details, refer to [3.2.4 Setting the Inspection Region](#).



[Extract Colors]

Sets the color extraction settings. [Color to Binary] is used for the extract color mode in this tool. Refer to [4.1 Extract Colors Settings](#) for the general setting method of [Color to Binary].



[Image Enhance]

Sets the image enhancement settings. No image enhancement is used in this tool.

[Detection Conditions]

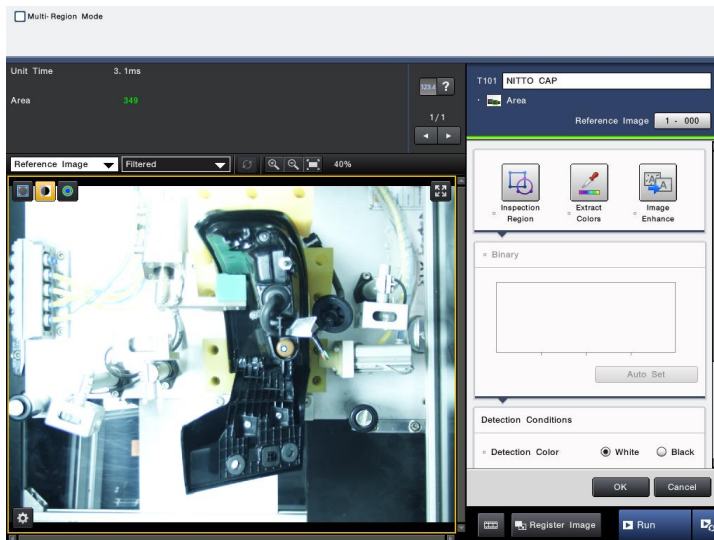
Sets the detection conditions according to the measurement items. For details, refer to [3.2.5 Adjustment for Detection Conditions](#).

[Judgment Conditions]

Sets the judgment conditions according to the measurement item. For details, refer to [3.2.6 Adjustment for Judgment Conditions](#).

3.2.4. Setting the Inspection Region

3.2.4.1. Inspection Region



Specifies the region to be measured.

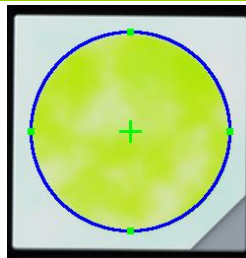
Circle is used for specifying the inspection region in this particular case.

[Circle]

(How to setup)

Specify three points on the target circle.

Drag the green points to change the size, and drag from the inside of the circle to move the location.



3.2.5. Adjustment for Detection Conditions

3.2.5.1. Detection Color

Detection Conditions

Detection Color

☒ White

☐ Black

Selects which area ([Black] or [White]) of the binarized image to measure.

(Reference) When "Color to Binary" mode is selected, the extracted color becomes [White].

[Detection Color]

(General Overview)

"Detection Color" specifies the color (either black or white) whose area will be measured in a binary image.



(Usage)

When the area to be measured is white, set this parameter to "White", and for black, set it to "Black".

Detection Color

☒ White

☐ Black

The area of the white part of the mold is found.

	Measured
Area	1097

Detection Color

☐ White

☒ Black

The area of the black part of the mold is found.

	Measured
Area	6803

3.2.6. Adjustment for Judgment Conditions

Judgment Conditions

Area

Upper Limit

Measured

349

Lower Limit

00000280

Set the range (upper/lower limit) judged as OK for the measured value.

The item(s) of judgment conditions set in this tool is/are as follows:

Area

The [Measured] field shows the measured result for the current image.

It is a useful reference for setting the upper/lower limit.

3.2.6.1. Defaults List of Judgment Conditions

Judgment Conditions Name	Defaults		On Change		Memo
	Upper Limit	Lower Limit	Upper Limit	Lower Limit	
Area	-----	280			

3.3. (CAM1) [T102]GASKET-1



3.3.1. Summary

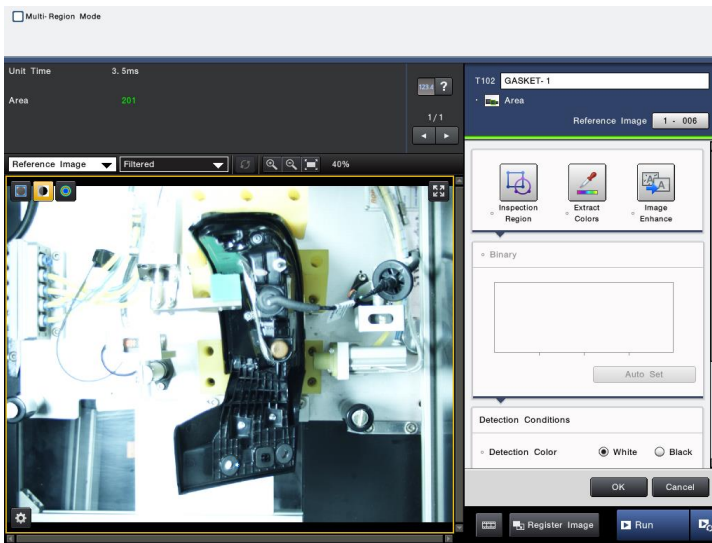
Summary of this tool
Measures the "White" or "Black" area after binarization (black-and-white processing) of the captured image.
Memo

3.3.2. Defaults List of Parameters

Parameter Name	Defaults	On Change	Memo
Detection Color	White		

3.3.3. Tool Edit Screen View

This is the edit screen for the [T102]GASKET-1 tool.



[Inspection Region]

Sets the inspection region. For details, refer to [3.3.4 Setting the Inspection Region](#).



[Extract Colors]

Sets the color extraction settings. [Color to Binary] is used for the extract color mode in this tool. Refer to [4.1 Extract Colors Settings](#) for the general setting method of [Color to Binary].



[Image Enhance]

Sets the image enhancement settings. No image enhancement is used in this tool.

[Detection Conditions]

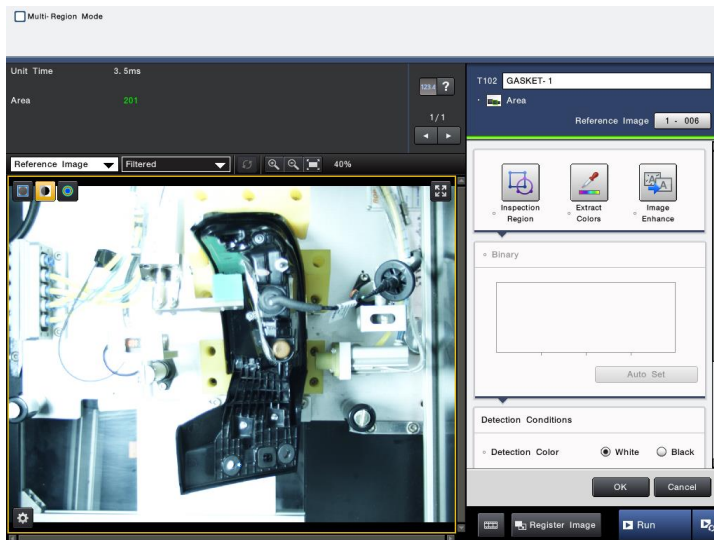
Sets the detection conditions according to the measurement items. For details, refer to [3.3.5 Adjustment for Detection Conditions](#).

[Judgment Conditions]

Sets the judgment conditions according to the measurement item. For details, refer to [3.3.6 Adjustment for Judgment Conditions](#).

3.3.4. Setting the Inspection Region

3.3.4.1. Inspection Region



Specifies the region to be measured.

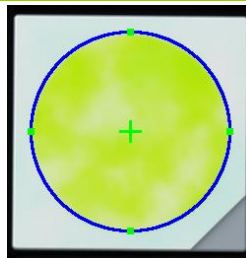
Circle is used for specifying the inspection region in this particular case.

[Circle]

(How to setup)

Specify three points on the target circle.

Drag the green points to change the size, and drag from the inside of the circle to move the location.



3.3.5. Adjustment for Detection Conditions

3.3.5.1. Detection Color

Detection Conditions	
◦ Detection Color	☒ White ☐ Black

Selects which area ([Black] or [White]) of the binarized image to measure.

(Reference) When "Color to Binary" mode is selected, the extracted color becomes [White].

[Detection Color]

(General Overview)

"Detection Color" specifies the color (either black or white) whose area will be measured in a binary image.



(Usage)

When the area to be measured is white, set this parameter to "White", and for black, set it to "Black".

Detection Color	
	☒ White ☐ Black

The area of the white part of the mold is found.

Measured	
Area	1097

Detection Color	
	☐ White ☒ Black

The area of the black part of the mold is found.

Measured	
Area	6803

3.3.6. Adjustment for Judgment Conditions

Judgment Conditions

Area

Upper Limit

Measured

201

Lower Limit

00000060

Set the range (upper/lower limit) judged as OK for the measured value.

The item(s) of judgment conditions set in this tool is/are as follows:

Area

The [Measured] field shows the measured result for the current image.

It is a useful reference for setting the upper/lower limit.

3.3.6.1. Defaults List of Judgment Conditions

Judgment Conditions Name	Defaults		On Change		Memo
	Upper Limit	Lower Limit	Upper Limit	Lower Limit	
Area	-----	60			

3.4. (CAM1) [T103]GASKET-2



3.4.1. Summary

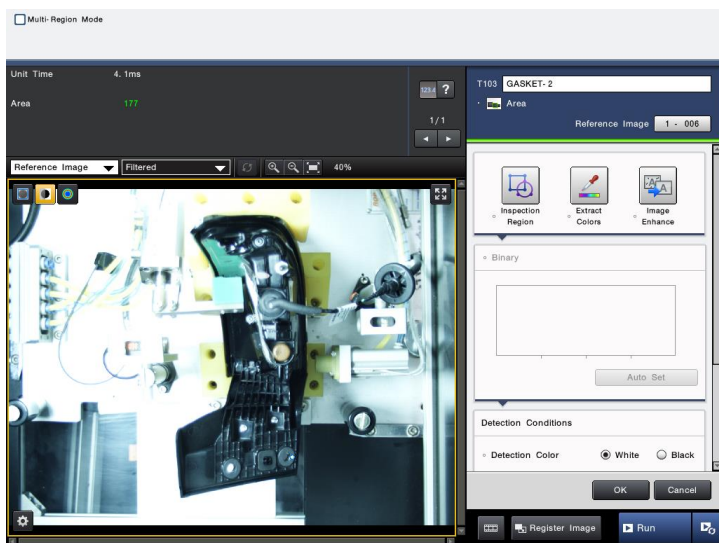
Summary of this tool
Measures the "White" or "Black" area after binarization (black-and-white processing) of the captured image.
Memo

3.4.2. Defaults List of Parameters

Parameter Name	Defaults	On Change	Memo
Detection Color	White		

3.4.3. Tool Edit Screen View

This is the edit screen for the [T103]GASKET-2 tool.



[Inspection Region]

Sets the inspection region. For details, refer to [3.4.4 Setting the Inspection Region](#).



[Extract Colors]

Sets the color extraction settings. [Color to Binary] is used for the extract color mode in this tool. Refer to [4.1 Extract Colors Settings](#) for the general setting method of [Color to Binary].



[Image Enhance]

Sets the image enhancement settings. No image enhancement is used in this tool.

[Detection Conditions]

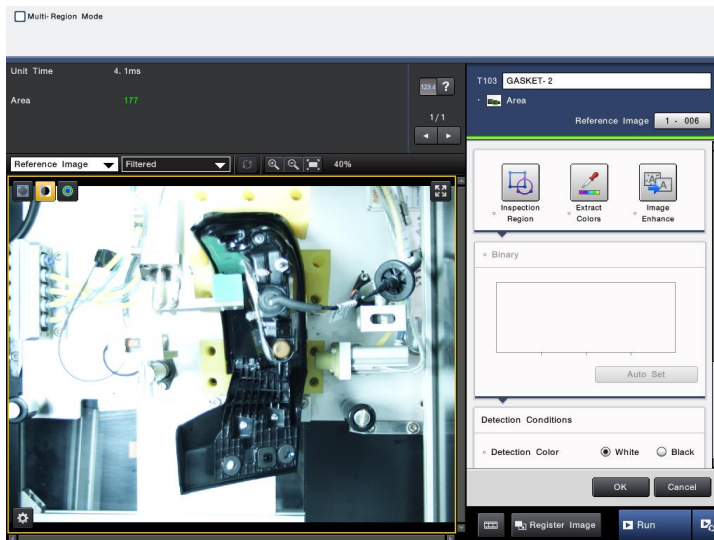
Sets the detection conditions according to the measurement items. For details, refer to [3.4.5 Adjustment for Detection Conditions](#).

[Judgment Conditions]

Sets the judgment conditions according to the measurement item. For details, refer to [3.4.6 Adjustment for Judgment Conditions](#).

3.4.4. Setting the Inspection Region

3.4.4.1. Inspection Region



Specifies the region to be measured.

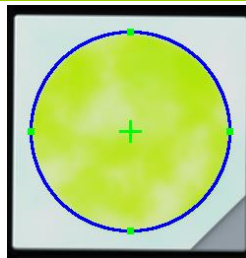
Circle is used for specifying the inspection region in this particular case.

[Circle]

(How to setup)

Specify three points on the target circle.

Drag the green points to change the size, and drag from the inside of the circle to move the location.



3.4.5. Adjustment for Detection Conditions

3.4.5.1. Detection Color

Detection Conditions

Detection Color

☒ White

☐ Black

Selects which area ([Black] or [White]) of the binarized image to measure.

(Reference) When "Color to Binary" mode is selected, the extracted color becomes [White].

[Detection Color]

(General Overview)

“Detection Color” specifies the color (either black or white) whose area will be measured in a binary image.



(Usage)

When the area to be measured is white, set this parameter to “White”, and for black, set it to “Black”.

Detection Color

☒ White

☐ Black

The area of the white part of the mold is found.

Measured	
Area	1097

Detection Color

☐ White

☒ Black

The area of the black part of the mold is found.

Measured	
Area	6803

3.4.6. Adjustment for Judgment Conditions

Judgment Conditions	
◦ Area	
Upper Limit	-----
Measured	177
Lower Limit	00000060

Set the range (upper/lower limit) judged as OK for the measured value.

The item(s) of judgment conditions set in this tool is/are as follows:

Area

The [Measured] field shows the measured result for the current image.

It is a useful reference for setting the upper/lower limit.

3.4.6.1. Defaults List of Judgment Conditions

Judgment Conditions Name	Defaults		On Change		Memo
	Upper Limit	Lower Limit	Upper Limit	Lower Limit	
Area	-----	60			

3.5. (CAM1) [T104]SQUARE GASKET



3.5.1. Summary

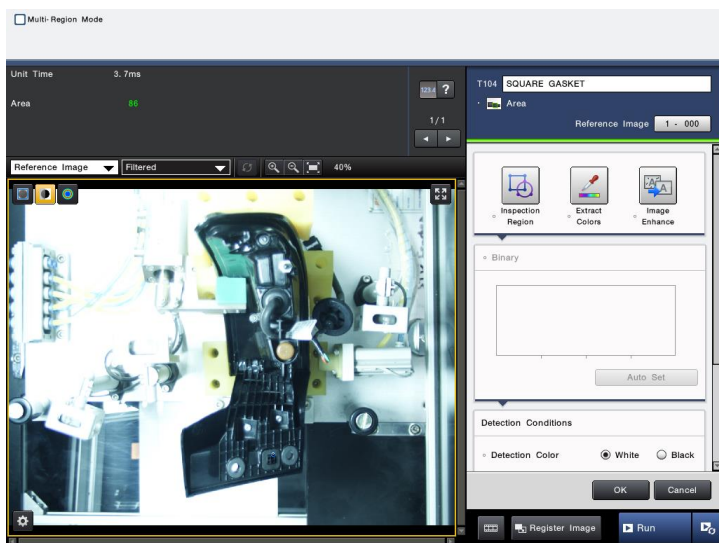
Summary of this tool
Measures the "White" or "Black" area after binarization (black-and-white processing) of the captured image.
Memo

3.5.2. Defaults List of Parameters

Parameter Name	Defaults	On Change	Memo
Detection Color	White		

3.5.3. Tool Edit Screen View

This is the edit screen for the [T104]SQUARE GASKET tool.



[Inspection Region]

Sets the inspection region. For details, refer to [3.5.4 Setting the Inspection Region](#).



[Extract Colors]

Sets the color extraction settings. [Color to Binary] is used for the extract color mode in this tool. Refer to [4.1 Extract Colors Settings](#) for the general setting method of [Color to Binary].



[Image Enhance]

Sets the image enhancement settings. No image enhancement is used in this tool.

[Detection Conditions]

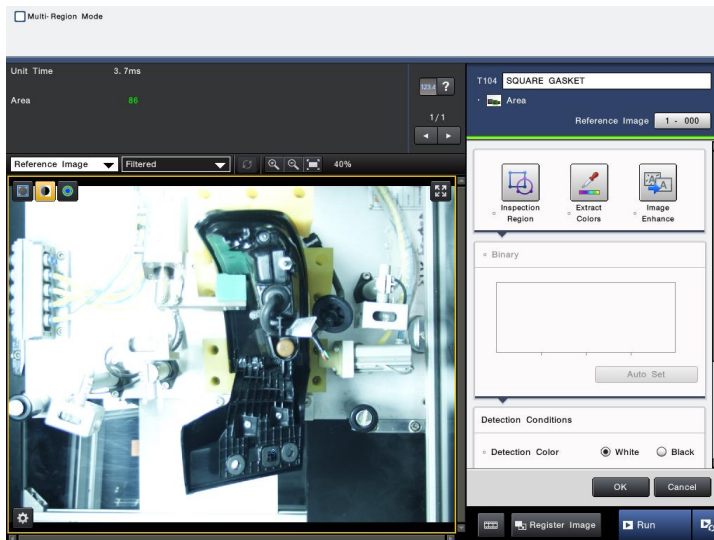
Sets the detection conditions according to the measurement items. For details, refer to [3.5.5 Adjustment for Detection Conditions](#).

[Judgment Conditions]

Sets the judgment conditions according to the measurement item. For details, refer to [3.5.6 Adjustment for Judgment Conditions](#).

3.5.4. Setting the Inspection Region

3.5.4.1. Inspection Region



Specifies the region to be measured.

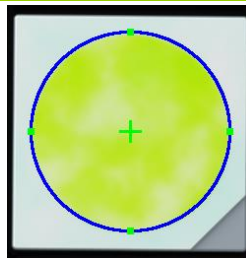
Circle is used for specifying the inspection region in this particular case.

[Circle]

(How to setup)

Specify three points on the target circle.

Drag the green points to change the size, and drag from the inside of the circle to move the location.



3.5.5. Adjustment for Detection Conditions

3.5.5.1. Detection Color

Detection Conditions

Detection Color

☒ White

☐ Black

Selects which area ([Black] or [White]) of the binarized image to measure.

(Reference) When "Color to Binary" mode is selected, the extracted color becomes [White].

[Detection Color]

(General Overview)

“Detection Color” specifies the color (either black or white) whose area will be measured in a binary image.



(Usage)

When the area to be measured is white, set this parameter to “White”, and for black, set it to “Black”.

Detection Color

☒ White

☐ Black

The area of the white part of the mold is found.

Measured	
Area	1097

Detection Color

☐ White

☒ Black

The area of the black part of the mold is found.

Measured	
Area	6803

3.5.6. Adjustment for Judgment Conditions

Judgment Conditions

Area

Upper Limit

Measured

86

Lower Limit

00000050

Set the range (upper/lower limit) judged as OK for the measured value.

The item(s) of judgment conditions set in this tool is/are as follows:

Area

The [Measured] field shows the measured result for the current image.

It is a useful reference for setting the upper/lower limit.

3.5.6.1. Defaults List of Judgment Conditions

Judgment Conditions Name	Defaults		On Change		Memo
	Upper Limit	Lower Limit	Upper Limit	Lower Limit	
Area	-----	50			

3.6. (CAM1) [T105]FINISHER



3.6.1. Summary

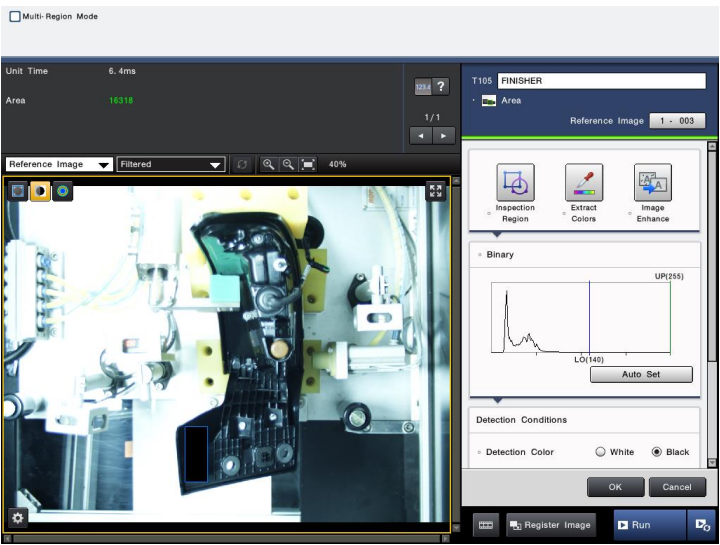
Summary of this tool
Measures the "White" or "Black" area after binarization (black-and-white processing) of the captured image.
Memo

3.6.2. Defaults List of Parameters

Parameter Name	Defaults	On Change	Memo
Upper Binary Threshold	255		
Lower Binary Threshold	140		
Detection Color	Black		

3.6.3. Tool Edit Screen View

This is the edit screen for the [T105]FINISHER tool.



[Inspection Region]

Sets the inspection region. For details, refer to [3.6.4 Setting the Inspection Region](#).



[Extract Colors]

Sets the color extraction settings. [Gray] is used for the extract color mode in this tool.



[Image Enhance]

Sets the image enhancement settings. The type(s) of Image Enhance used in this tool

is/are as follows. Refer to [4.2 Image Enhance Settings](#) for the summary of each Image Enhance type.

Image Enhance List

0 Binary

[Binary]

Sets the binary settings. For details, refer to [3.6.5 Adjustment for Binary](#).

[Detection Conditions]

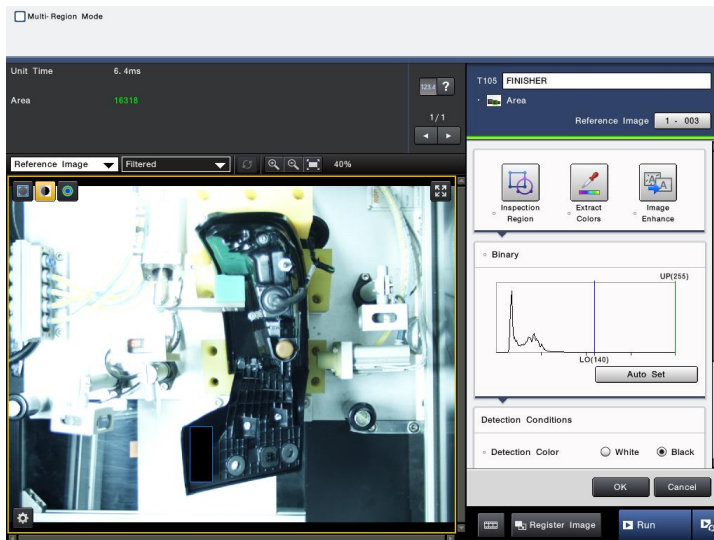
Sets the detection conditions according to the measurement items. For details, refer to [3.6.6 Adjustment for Detection Conditions](#).

[Judgment Conditions]

Sets the judgment conditions according to the measurement item. For details, refer to [3.6.7 Adjustment for Judgment Conditions](#).

3.6.4. Setting the Inspection Region

3.6.4.1. Inspection Region



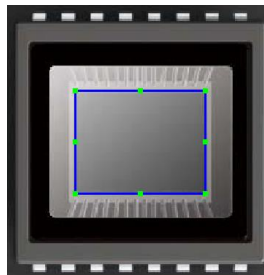
Specifies the region to be measured.

Rectangle is used for specifying the inspection region in this particular case.

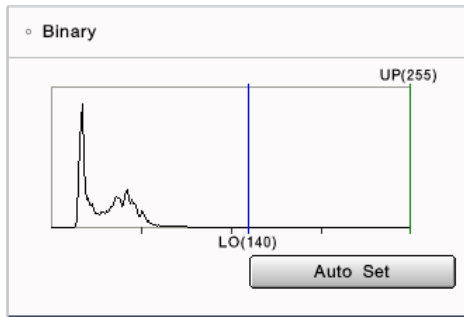
[Rectangle]

(How to setup)

Drag each green point to adjust the size and position. When adjusting only the position without changing the size, drag from the inside of the frame border.



3.6.5. Adjustment for Binary

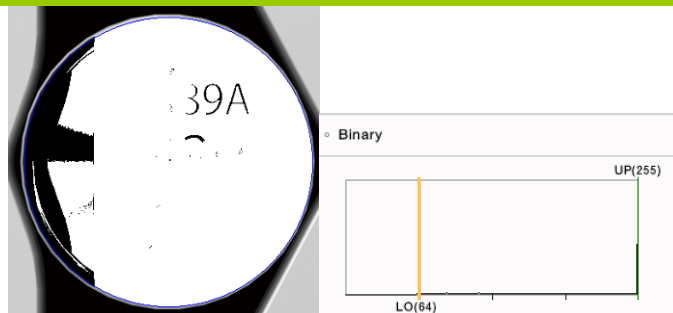


Specifies the threshold value of the upper (UP) and lower (LO) limit on the histogram showing the concentration distribution in the inspection region to binarize the image as follows: The pixels having a intensity within the range of the threshold are converted to white and other pixels to black.

[Binary]

(General Overview)

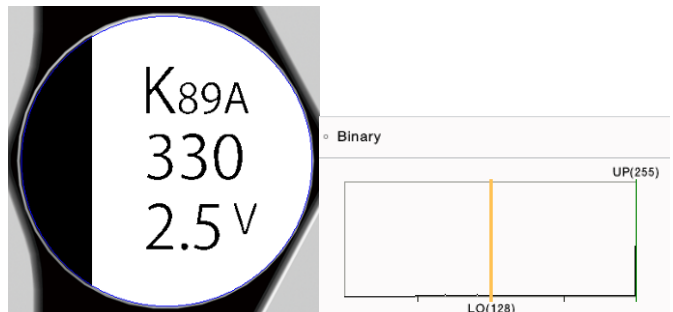
“Binary” is a filter which binarizes an image to white (255 tones) and black (0 tone). This filter is mainly used in “Area” and “Blob” tools.



(Usage)

↓ Change the LO value from 64 to 128.

Basically, change the “LO” value to set the range where brightness is converted to white (255 tones) and the range where brightness is converted to black (0 tone).



“LO/UP”: Sets brightness falling in the range between “LO” and “UP” to white (255 tones), and brightness out of the range to black (0 tone).

The image is changed after use of “Binary” filter.

3.6.6. Adjustment for Detection Conditions

3.6.6.1. Detection Color

Detection Conditions	
◦ Detection Color	☐ White ☒ Black

Selects which area ([Black] or [White]) of the binarized image to measure.

(Reference) When "Color to Binary" mode is selected, the extracted color becomes [White].

[Detection Color]

(General Overview)

“Detection Color” specifies the color (either black or white) whose area will be measured in a binary image.



(Usage)

When the area to be measured is white, set this parameter to “White”, and for black, set it to “Black”.

Detection Color

☒ White ☐ Black

The area of the white part of the mold is found.

	Measured
Area	1097

◦ Detection Color

☐ White ☒ Black

The area of the black part of the mold is found.

	Measured
Area	6803

3.6.7. Adjustment for Judgment Conditions

Judgment Conditions	
◦ Area	
Upper Limit	-----
Measured	16318
Lower Limit	00013000

Set the range (upper/lower limit) judged as OK for the measured value.

The item(s) of judgment conditions set in this tool is/are as follows:

Area

The [Measured] field shows the measured result for the current image.

It is a useful reference for setting the upper/lower limit.

3.6.7.1. Defaults List of Judgment Conditions

Judgment Conditions Name	Defaults		On Change		Memo
	Upper Limit	Lower Limit	Upper Limit	Lower Limit	
Area	-----	13000			

3.7. (CAM1) [T106]SCREW-1



3.7.1. Summary

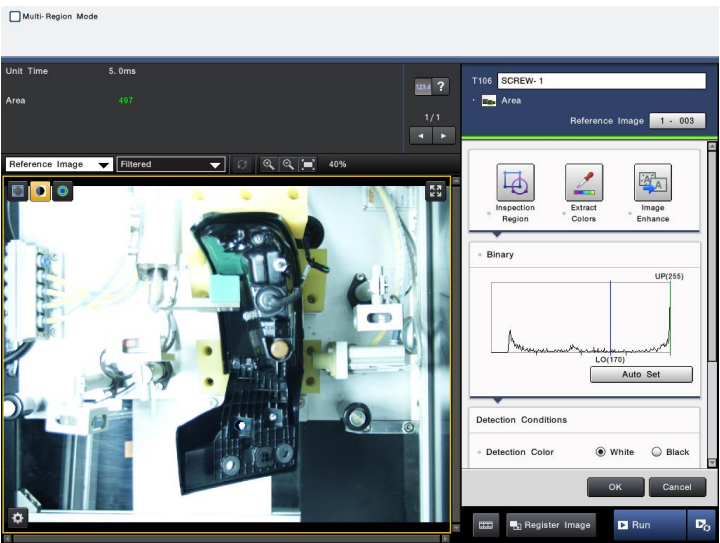
Summary of this tool
Measures the "White" or "Black" area after binarization (black-and-white processing) of the captured image.
Memo

3.7.2. Defaults List of Parameters

Parameter Name	Defaults	On Change	Memo
Upper Binary Threshold	255		
Lower Binary Threshold	170		
Detection Color	White		

3.7.3. Tool Edit Screen View

This is the edit screen for the [T106]SCREW-1 tool.



[Inspection Region]

Sets the inspection region. For details, refer to [3.7.4 Setting the Inspection Region](#).



[Extract Colors]

Sets the color extraction settings. [Gray] is used for the extract color mode in this tool.



[Image Enhance]

Sets the image enhancement settings. The type(s) of Image Enhance used in this tool

is/are as follows. Refer to [4.2 Image Enhance Settings](#) for the summary of each Image Enhance type.

Image Enhance List

0 Binary

[Binary]

Sets the binary settings. For details, refer to [3.7.5 Adjustment for Binary](#).

[Detection Conditions]

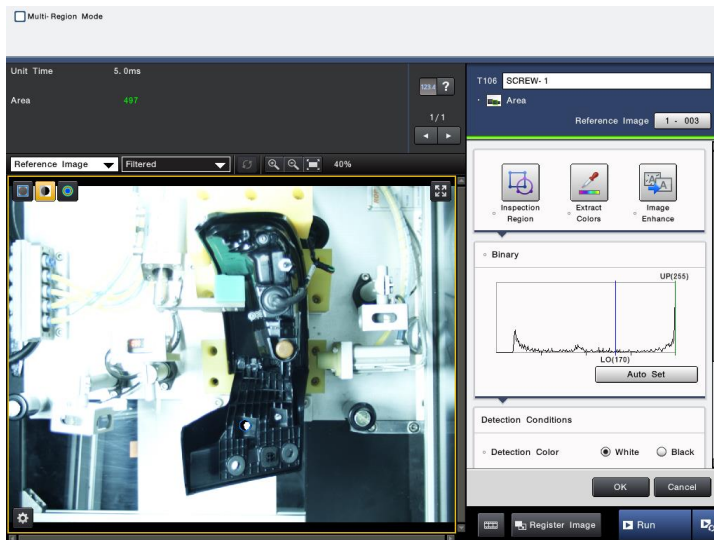
Sets the detection conditions according to the measurement items. For details, refer to [3.7.6 Adjustment for Detection Conditions](#).

[Judgment Conditions]

Sets the judgment conditions according to the measurement item. For details, refer to [3.7.7 Adjustment for Judgment Conditions](#).

3.7.4. Setting the Inspection Region

3.7.4.1. Inspection Region



Specifies the region to be measured.

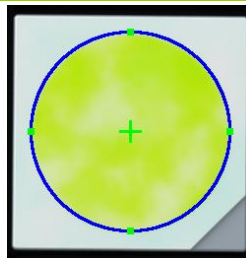
Circle is used for specifying the inspection region in this particular case.

[Circle]

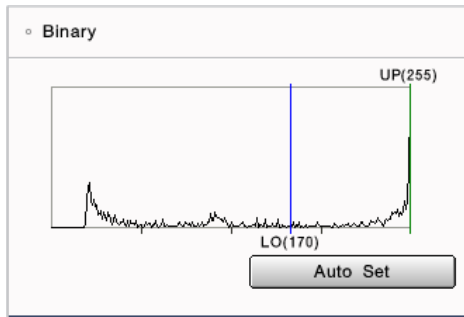
(How to setup)

Specify three points on the target circle.

Drag the green points to change the size, and drag from the inside of the circle to move the location.



3.7.5. Adjustment for Binary

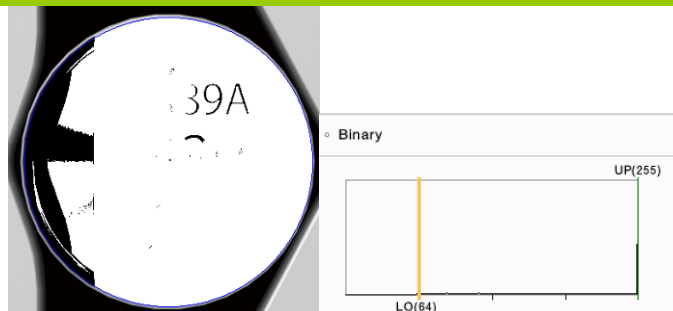


Specifies the threshold value of the upper (UP) and lower (LO) limit on the histogram showing the concentration distribution in the inspection region to binarize the image as follows: The pixels having a intensity within the range of the threshold are converted to white and other pixels to black.

[Binary]

(General Overview)

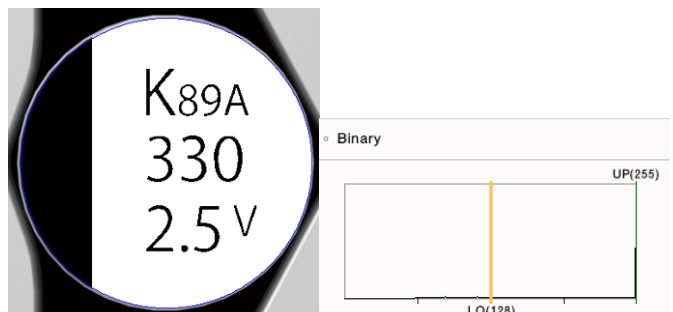
“Binary” is a filter which binarizes an image to white (255 tones) and black (0 tone). This filter is mainly used in “Area” and “Blob” tools.



(Usage)

↓ Change the LO value from 64 to 128.

Basically, change the “LO” value to set the range where brightness is converted to white (255 tones) and the range where brightness is converted to black (0 tone).



“LO/UP”: Sets brightness falling in the range between “LO” and “UP” to white (255 tones), and brightness out of the range to black (0 tone).

The image is changed after use of “Binary” filter.

3.7.6. Adjustment for Detection Conditions

3.7.6.1. Detection Color

Detection Conditions

Detection Color

☒ White

☐ Black

Selects which area ([Black] or [White]) of the binarized image to measure.

(Reference) When "Color to Binary" mode is selected, the extracted color becomes [White].

[Detection Color]

(General Overview)

“Detection Color” specifies the color (either black or white) whose area will be measured in a binary image.



(Usage)

When the area to be measured is white, set this parameter to “White”, and for black, set it to “Black”.

Detection Color

☒ White

☐ Black

The area of the white part of the mold is found.

	Measured
Area	1097

Detection Color

☐ White

☒ Black

The area of the black part of the mold is found.

	Measured
Area	6803

3.7.7. Adjustment for Judgment Conditions

Judgment Conditions

Area

Upper Limit

Measured

497

Lower Limit

00000324

Set the range (upper/lower limit) judged as OK for the measured value.

The item(s) of judgment conditions set in this tool is/are as follows:

Area

The [Measured] field shows the measured result for the current image.

It is a useful reference for setting the upper/lower limit.

3.7.7.1. Defaults List of Judgment Conditions

Judgment Conditions Name	Defaults		On Change		Memo
	Upper Limit	Lower Limit	Upper Limit	Lower Limit	
Area	-----	324			

3.8. (CAM1) [T107]SCREW-2



3.8.1. Summary

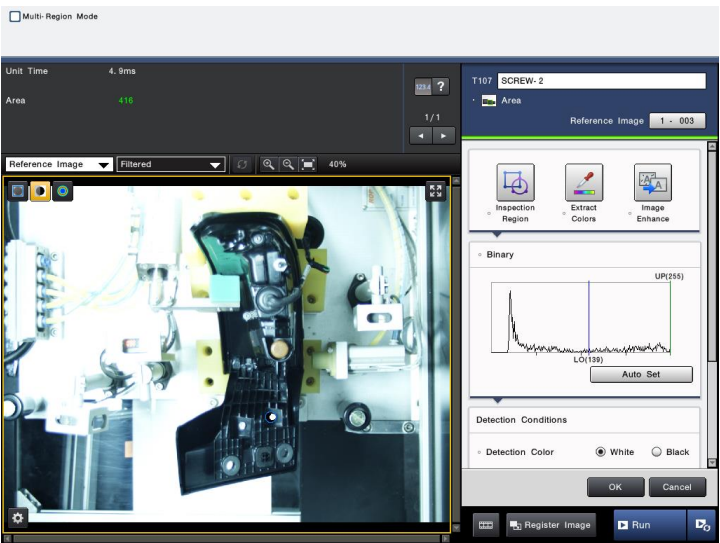
Summary of this tool
Measures the "White" or "Black" area after binarization (black-and-white processing) of the captured image.
Memo

3.8.2. Defaults List of Parameters

Parameter Name	Defaults	On Change	Memo
Upper Binary Threshold	255		
Lower Binary Threshold	139		
Detection Color	White		

3.8.3. Tool Edit Screen View

This is the edit screen for the [T107]SCREW-2 tool.



[Inspection Region]

Sets the inspection region. For details, refer to [3.8.4 Setting the Inspection Region](#).



[Extract Colors]

Sets the color extraction settings. [Gray] is used for the extract color mode in this tool.



[Image Enhance]

Sets the image enhancement settings. The type(s) of Image Enhance used in this tool

is/are as follows. Refer to [4.2 Image Enhance Settings](#) for the summary of each Image Enhance type.

Image Enhance List
0 Binary

[Binary]

Sets the binary settings. For details, refer to [3.8.5 Adjustment for Binary](#).

[Detection Conditions]

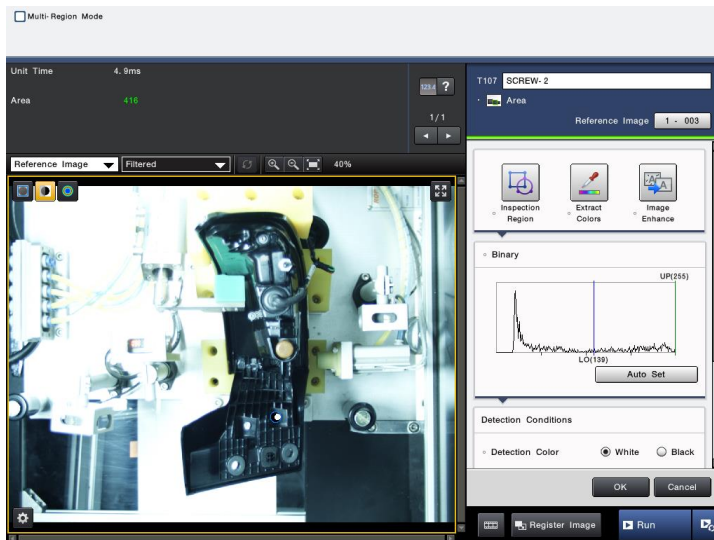
Sets the detection conditions according to the measurement items. For details, refer to [3.8.6 Adjustment for Detection Conditions](#).

[Judgment Conditions]

Sets the judgment conditions according to the measurement item. For details, refer to [3.8.7 Adjustment for Judgment Conditions](#).

3.8.4. Setting the Inspection Region

3.8.4.1. Inspection Region



Specifies the region to be measured.

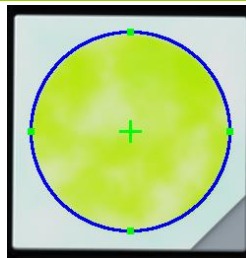
Circle is used for specifying the inspection region in this particular case.

[Circle]

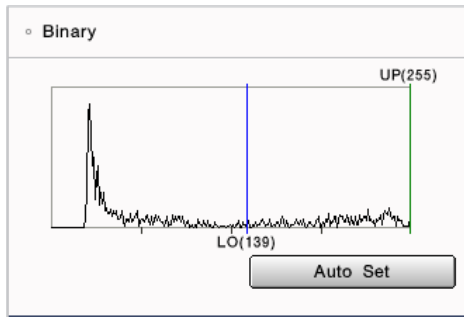
(How to setup)

Specify three points on the target circle.

Drag the green points to change the size, and drag from the inside of the circle to move the location.



3.8.5. Adjustment for Binary

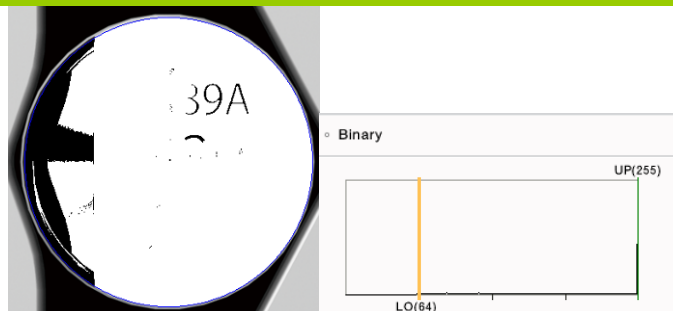


Specifies the threshold value of the upper (UP) and lower (LO) limit on the histogram showing the concentration distribution in the inspection region to binarize the image as follows: The pixels having a intensity within the range of the threshold are converted to white and other pixels to black.

[Binary]

(General Overview)

“Binary” is a filter which binarizes an image to white (255 tones) and black (0 tone). This filter is mainly used in “Area” and “Blob” tools.

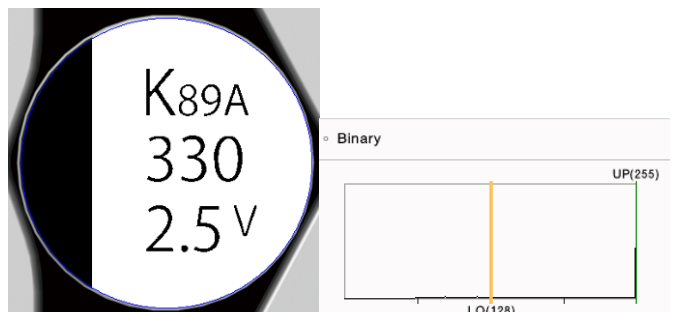


(Usage)

↓ Change the LO value from 64 to 128.

Basically, change the “LO” value to set the range where brightness is converted to white (255 tones) and the range where brightness is converted to black (0 tone).

“LO/UP”: Sets brightness falling in the range between “LO” and “UP” to white (255 tones), and brightness out of the range to black (0 tone).



The image is changed after use of “Binary” filter.

3.8.6. Adjustment for Detection Conditions

3.8.6.1. Detection Color

Detection Conditions	
◦ Detection Color	☒ White ☐ Black

Selects which area ([Black] or [White]) of the binarized image to measure.

(Reference) When "Color to Binary" mode is selected, the extracted color becomes [White].

[Detection Color]

(General Overview)

"Detection Color" specifies the color (either black or white) whose area will be measured in a binary image.



(Usage)

When the area to be measured is white, set this parameter to "White", and for black, set it to "Black".

Detection Color	
	☒ White ☐ Black

The area of the white part of the mold is found.

Measured	
Area	1097

Detection Color	
	☐ White ☒ Black

The area of the black part of the mold is found.

Measured	
Area	6803

3.8.7. Adjustment for Judgment Conditions

Judgment Conditions	
◦ Area	
Upper Limit	-----
Measured	416
Lower Limit	00000228

Set the range (upper/lower limit) judged as OK for the measured value.

The item(s) of judgment conditions set in this tool is/are as follows:

Area

The [Measured] field shows the measured result for the current image.

It is a useful reference for setting the upper/lower limit.

3.8.7.1. Defaults List of Judgment Conditions

Judgment Conditions Name	Defaults		On Change		Memo
	Upper Limit	Lower Limit	Upper Limit	Lower Limit	
Area	-----	228			

3.9. (CAM2) [T201]NITTO CAP



3.9.1. Summary

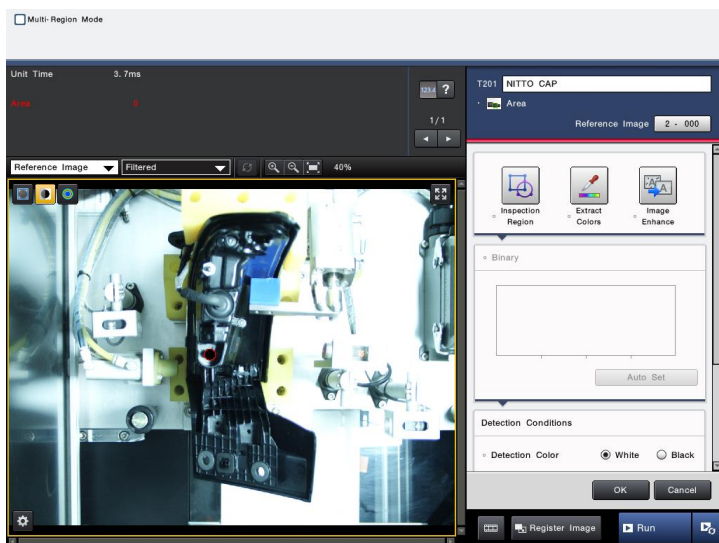
Summary of this tool
Measures the "White" or "Black" area after binarization (black-and-white processing) of the captured image.
Memo

3.9.2. Defaults List of Parameters

Parameter Name	Defaults	On Change	Memo
Detection Color	White		

3.9.3. Tool Edit Screen View

This is the edit screen for the [T201]NITTO CAP tool.



[Inspection Region]

Sets the inspection region. For details, refer to [3.9.4 Setting the Inspection Region](#).



[Extract Colors]

Sets the color extraction settings. [Color to Binary] is used for the extract color mode in this tool. Refer to [4.1 Extract Colors Settings](#) for the general setting method of [Color to Binary].



[Image Enhance]

Sets the image enhancement settings. No image enhancement is used in this tool.

[Detection Conditions]

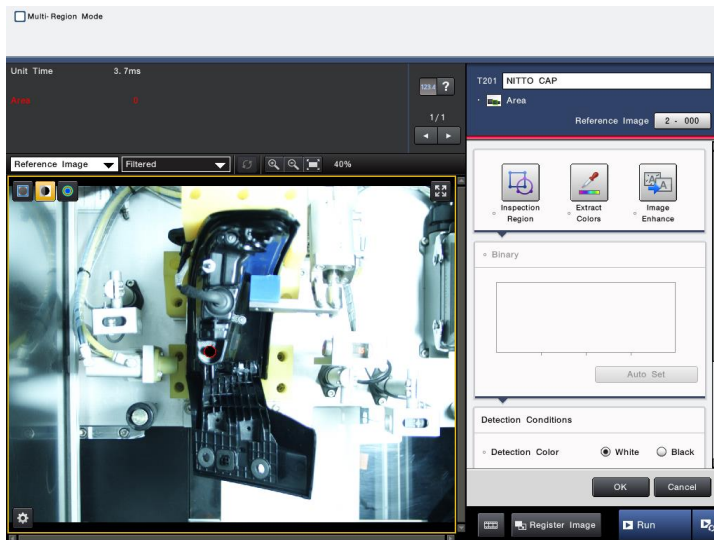
Sets the detection conditions according to the measurement items. For details, refer to [3.9.5 Adjustment for Detection Conditions](#).

[Judgment Conditions]

Sets the judgment conditions according to the measurement item. For details, refer to [3.9.6 Adjustment for Judgment Conditions](#).

3.9.4. Setting the Inspection Region

3.9.4.1. Inspection Region



Specifies the region to be measured.

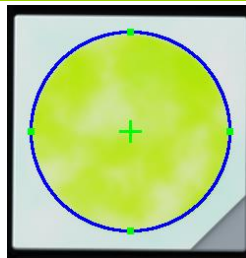
Circle is used for specifying the inspection region in this particular case.

[Circle]

(How to setup)

Specify three points on the target circle.

Drag the green points to change the size, and drag from the inside of the circle to move the location.



3.9.5. Adjustment for Detection Conditions

3.9.5.1. Detection Color

Detection Conditions

Detection Color

☒ White

☐ Black

Selects which area ([Black] or [White]) of the binarized image to measure.

(Reference) When "Color to Binary" mode is selected, the extracted color becomes [White].

[Detection Color]

(General Overview)

“Detection Color” specifies the color (either black or white) whose area will be measured in a binary image.



(Usage)

When the area to be measured is white, set this parameter to “White”, and for black, set it to “Black”.

Detection Color

☒ White

☐ Black

The area of the white part of the mold is found.

	Measured
Area	1097

Detection Color

☐ White

☒ Black

The area of the black part of the mold is found.

	Measured
Area	6803

3.9.6. Adjustment for Judgment Conditions

Judgment Conditions

Area

Upper Limit

Measured

0

Lower Limit

00000574

Set the range (upper/lower limit) judged as OK for the measured value.

The item(s) of judgment conditions set in this tool is/are as follows:

Area

The [Measured] field shows the measured result for the current image.

It is a useful reference for setting the upper/lower limit.

3.9.6.1. Defaults List of Judgment Conditions

Judgment Conditions Name	Defaults		On Change		Memo
	Upper Limit	Lower Limit	Upper Limit	Lower Limit	
Area	-----	574			

3.10. (CAM2) [T202]GASKET-1



3.10.1. Summary

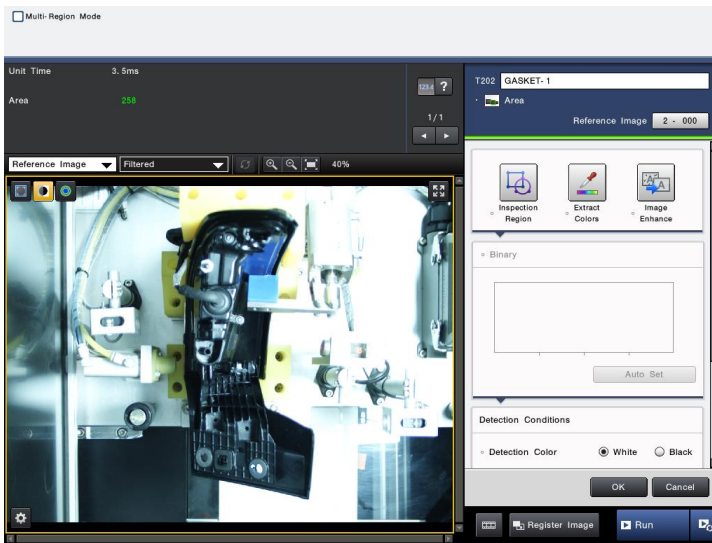
Summary of this tool
Measures the "White" or "Black" area after binarization (black-and-white processing) of the captured image.
Memo

3.10.2. Defaults List of Parameters

Parameter Name	Defaults	On Change	Memo
Detection Color	White		

3.10.3. Tool Edit Screen View

This is the edit screen for the [T202]GASKET-1 tool.



[Inspection Region]

Sets the inspection region. For details, refer to [3.10.4 Setting the Inspection Region](#).



[Extract Colors]

Sets the color extraction settings. [Color to Binary] is used for the extract color mode in this tool. Refer to [4.1 Extract Colors Settings](#) for the general setting method of [Color to Binary].



[Image Enhance]

Sets the image enhancement settings. No image enhancement is used in this tool.

[Detection Conditions]

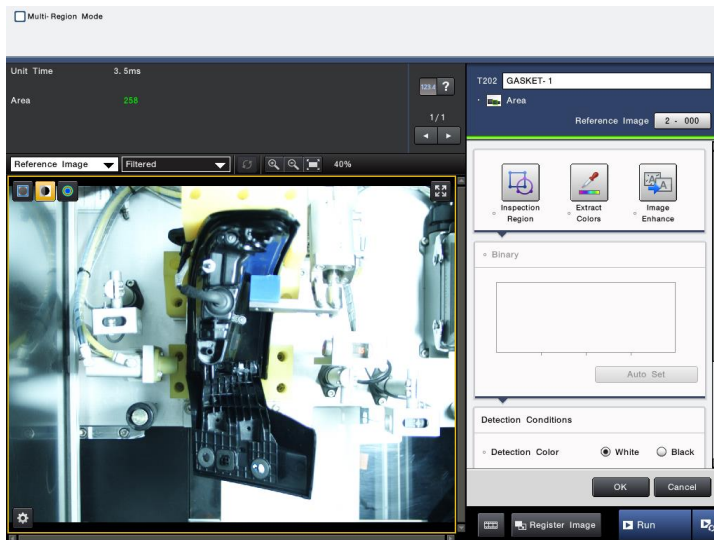
Sets the detection conditions according to the measurement items. For details, refer to [3.10.5 Adjustment for Detection Conditions](#).

[Judgment Conditions]

Sets the judgment conditions according to the measurement item. For details, refer to [3.10.6 Adjustment for Judgment Conditions](#).

3.10.4. Setting the Inspection Region

3.10.4.1. Inspection Region



Specifies the region to be measured.

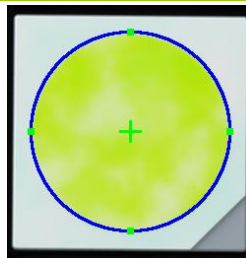
Circle is used for specifying the inspection region in this particular case.

[Circle]

(How to setup)

Specify three points on the target circle.

Drag the green points to change the size, and drag from the inside of the circle to move the location.



3.10.5. Adjustment for Detection Conditions

3.10.5.1. Detection Color

Detection Conditions

Detection Color

☒ White

☐ Black

Selects which area ([Black] or [White]) of the binarized image to measure.

(Reference) When "Color to Binary" mode is selected, the extracted color becomes [White].

[Detection Color]

(General Overview)

"Detection Color" specifies the color (either black or white) whose area will be measured in a binary image.



(Usage)

When the area to be measured is white, set this parameter to "White", and for black, set it to "Black".

Detection Color

☒ White

☐ Black

The area of the white part of the mold is found.

	Measured
Area	1097

Detection Color

☐ White

☒ Black

The area of the black part of the mold is found.

	Measured
Area	6803

3.10.6. Adjustment for Judgment Conditions

Judgment Conditions

Area

Upper Limit

Measured

258

Lower Limit

00000088

Set the range (upper/lower limit) judged as OK for the measured value.

The item(s) of judgment conditions set in this tool is/are as follows:

Area

The [Measured] field shows the measured result for the current image.

It is a useful reference for setting the upper/lower limit.

3.10.6.1. Defaults List of Judgment Conditions

Judgment Conditions Name	Defaults		On Change		Memo
	Upper Limit	Lower Limit	Upper Limit	Lower Limit	
Area	-----	88			

3.11. (CAM2) [T203]GASKET-2



3.11.1. Summary

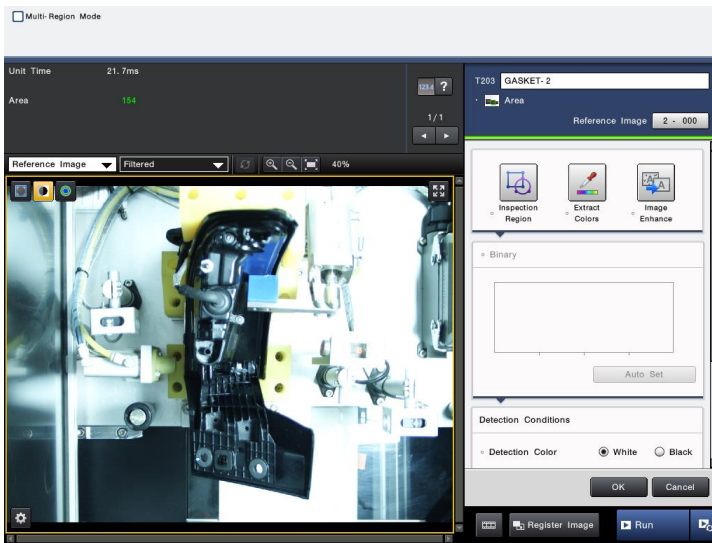
Summary of this tool
Measures the "White" or "Black" area after binarization (black-and-white processing) of the captured image.
Memo

3.11.2. Defaults List of Parameters

Parameter Name	Defaults	On Change	Memo
Detection Color	White		

3.11.3. Tool Edit Screen View

This is the edit screen for the [T203]GASKET-2 tool.



[Inspection Region]

Sets the inspection region. For details, refer to [3.11.4 Setting the Inspection Region](#).



[Extract Colors]

Sets the color extraction settings. [Color to Binary] is used for the extract color mode in this tool. Refer to [4.1 Extract Colors Settings](#) for the general setting method of [Color to Binary].



[Image Enhance]

Sets the image enhancement settings. No image enhancement is used in this tool.

[Detection Conditions]

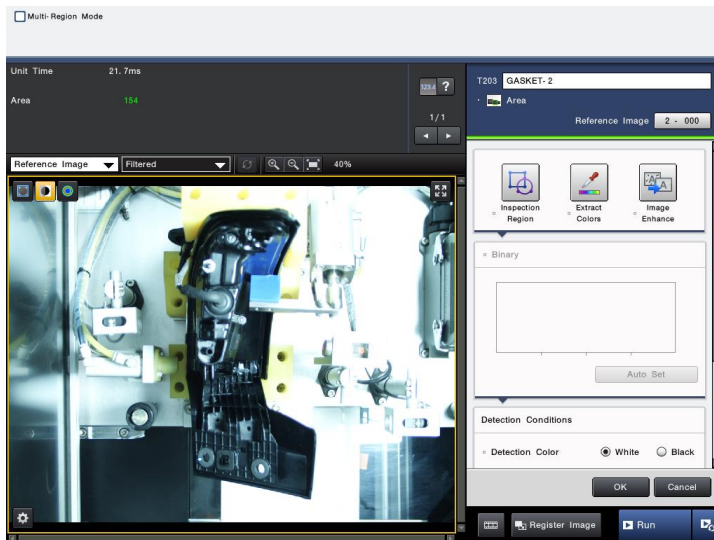
Sets the detection conditions according to the measurement items. For details, refer to [3.11.5 Adjustment for Detection Conditions](#).

[Judgment Conditions]

Sets the judgment conditions according to the measurement item. For details, refer to [3.11.6 Adjustment for Judgment Conditions](#).

3.11.4. Setting the Inspection Region

3.11.4.1. Inspection Region



Specifies the region to be measured.

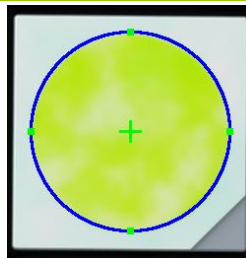
Circle is used for specifying the inspection region in this particular case.

[Circle]

(How to setup)

Specify three points on the target circle.

Drag the green points to change the size, and drag from the inside of the circle to move the location.



3.11.5. Adjustment for Detection Conditions

3.11.5.1. Detection Color

Detection Conditions	
◦ Detection Color	☒ White ☐ Black

Selects which area ([Black] or [White]) of the binarized image to measure.

(Reference) When "Color to Binary" mode is selected, the extracted color becomes [White].

[Detection Color]

(General Overview)

“Detection Color” specifies the color (either black or white) whose area will be measured in a binary image.



(Usage)

When the area to be measured is white, set this parameter to “White”, and for black, set it to “Black”.

Detection Color

☒ White ☐ Black

The area of the white part of the mold is found.

	Measured
Area	1097

◦ Detection Color

☐ White ☒ Black

The area of the black part of the mold is found.

	Measured
Area	6803

3.11.6. Adjustment for Judgment Conditions

Judgment Conditions

Area

Upper Limit

Measured

154

Lower Limit

00000061

Set the range (upper/lower limit) judged as OK for the measured value.

The item(s) of judgment conditions set in this tool is/are as follows:

Area

The [Measured] field shows the measured result for the current image.

It is a useful reference for setting the upper/lower limit.

3.11.6.1. Defaults List of Judgment Conditions

Judgment Conditions Name	Defaults		On Change		Memo
	Upper Limit	Lower Limit	Upper Limit	Lower Limit	
Area	-----	61			

3.12. (CAM2) [T204]SQUARE GASKET



3.12.1. Summary

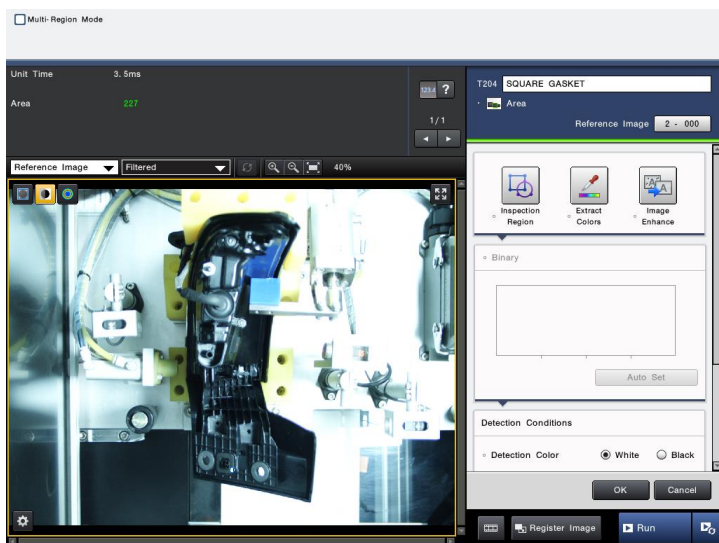
Summary of this tool
Measures the "White" or "Black" area after binarization (black-and-white processing) of the captured image.
Memo

3.12.2. Defaults List of Parameters

Parameter Name	Defaults	On Change	Memo
Detection Color	White		

3.12.3. Tool Edit Screen View

This is the edit screen for the [T204]SQUARE GASKET tool.



[Inspection Region]

Sets the inspection region. For details, refer to [3.12.4 Setting the Inspection Region](#).



[Extract Colors]

Sets the color extraction settings. [Color to Binary] is used for the extract color mode in this tool. Refer to [4.1 Extract Colors Settings](#) for the general setting method of [Color to Binary].



[Image Enhance]

Sets the image enhancement settings. No image enhancement is used in this tool.

[Detection Conditions]

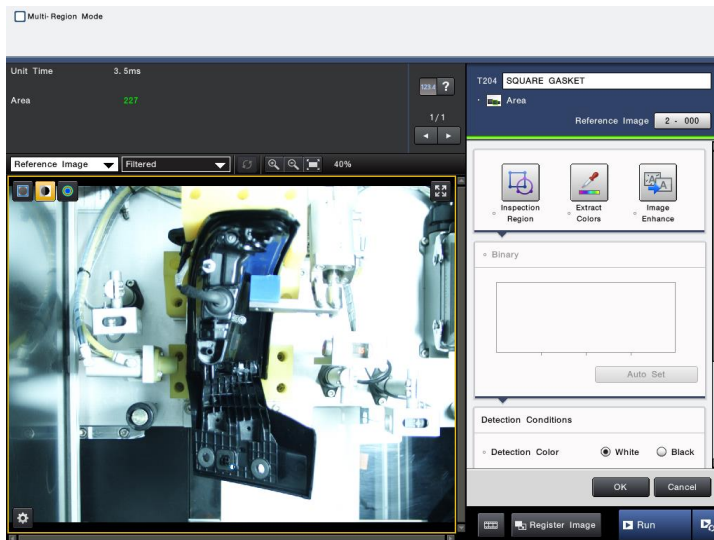
Sets the detection conditions according to the measurement items. For details, refer to [3.12.5 Adjustment for Detection Conditions](#).

[Judgment Conditions]

Sets the judgment conditions according to the measurement item. For details, refer to [3.12.6 Adjustment for Judgment Conditions](#).

3.12.4. Setting the Inspection Region

3.12.4.1. Inspection Region



Specifies the region to be measured.

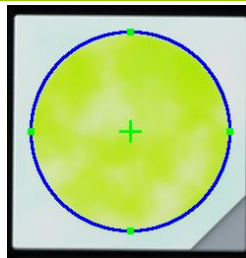
Circle is used for specifying the inspection region in this particular case.

[Circle]

(How to setup)

Specify three points on the target circle.

Drag the green points to change the size, and drag from the inside of the circle to move the location.



3.12.5. Adjustment for Detection Conditions

3.12.5.1. Detection Color

Detection Conditions

Detection Color

☒ White

☐ Black

Selects which area ([Black] or [White]) of the binarized image to measure.

(Reference) When "Color to Binary" mode is selected, the extracted color becomes [White].

[Detection Color]

(General Overview)

"Detection Color" specifies the color (either black or white) whose area will be measured in a binary image.



(Usage)

When the area to be measured is white, set this parameter to "White", and for black, set it to "Black".

Detection Color

☒ White

☐ Black

The area of the white part of the mold is found.

	Measured
Area	1097

Detection Color

☐ White

☒ Black

The area of the black part of the mold is found.

	Measured
Area	6803

3.12.6. Adjustment for Judgment Conditions

Judgment Conditions	
◦ Area	
Upper Limit	-----
Measured	227
Lower Limit	00000141

Set the range (upper/lower limit) judged as OK for the measured value.

The item(s) of judgment conditions set in this tool is/are as follows:

Area

The [Measured] field shows the measured result for the current image.

It is a useful reference for setting the upper/lower limit.

3.12.6.1. Defaults List of Judgment Conditions

Judgment Conditions Name	Defaults		On Change		Memo
	Upper Limit	Lower Limit	Upper Limit	Lower Limit	
Area	-----	141			

3.13. (CAM2) [T205]FINISHER



3.13.1. Summary

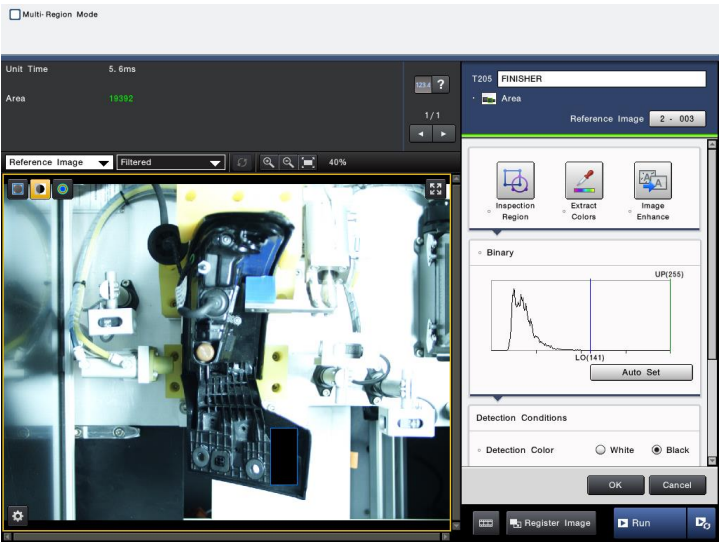
Summary of this tool
Measures the "White" or "Black" area after binarization (black-and-white processing) of the captured image.
Memo

3.13.2. Defaults List of Parameters

Parameter Name	Defaults	On Change	Memo
Upper Binary Threshold	255		
Lower Binary Threshold	141		
Detection Color	Black		

3.13.3. Tool Edit Screen View

This is the edit screen for the [T205]FINISHER tool.



[Inspection Region]

Sets the inspection region. For details, refer to [3.13.4 Setting the Inspection Region](#).



[Extract Colors]

Sets the color extraction settings. [Gray] is used for the extract color mode in this tool.



[Image Enhance]

Sets the image enhancement settings. The type(s) of Image Enhance used in this tool

is/are as follows. Refer to [4.2 Image Enhance Settings](#) for the summary of each Image Enhance type.

Image Enhance List

0 Binary

[Binary]

Sets the binary settings. For details, refer to [3.13.5 Adjustment for Binary](#).

[Detection Conditions]

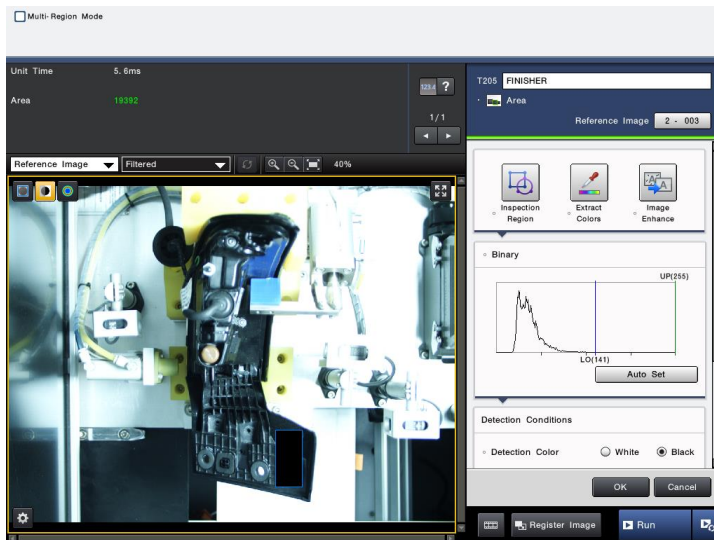
Sets the detection conditions according to the measurement items. For details, refer to [3.13.6 Adjustment for Detection Conditions](#).

[Judgment Conditions]

Sets the judgment conditions according to the measurement item. For details, refer to [3.13.7 Adjustment for Judgment Conditions](#).

3.13.4. Setting the Inspection Region

3.13.4.1. Inspection Region



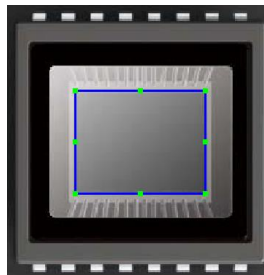
Specifies the region to be measured.

Rectangle is used for specifying the inspection region in this particular case.

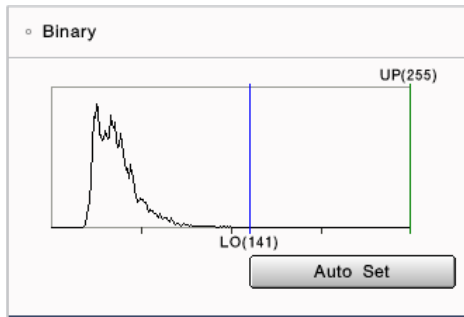
[Rectangle]

(How to setup)

Drag each green point to adjust the size and position. When adjusting only the position without changing the size, drag from the inside of the frame border.



3.13.5. Adjustment for Binary

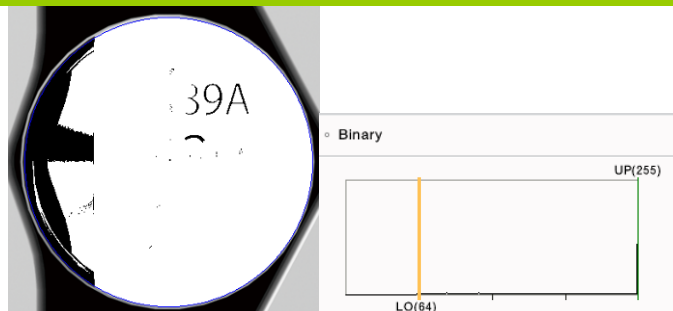


Specifies the threshold value of the upper (UP) and lower (LO) limit on the histogram showing the concentration distribution in the inspection region to binarize the image as follows: The pixels having a intensity within the range of the threshold are converted to white and other pixels to black.

[Binary]

(General Overview)

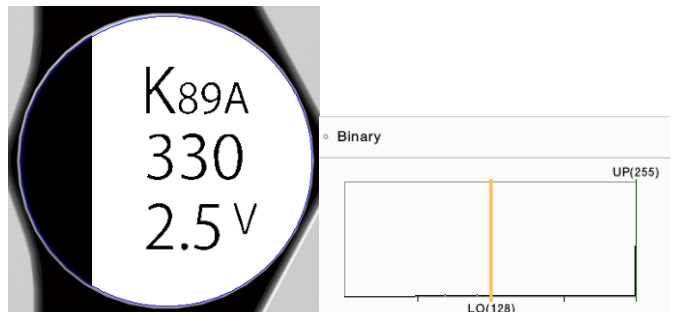
“Binary” is a filter which binarizes an image to white (255 tones) and black (0 tone). This filter is mainly used in “Area” and “Blob” tools.



(Usage)

↓ Change the LO value from 64 to 128.

Basically, change the “LO” value to set the range where brightness is converted to white (255 tones) and the range where brightness is converted to black (0 tone).



“LO/UP”: Sets brightness falling in the range between “LO” and “UP” to white (255 tones), and brightness out of the range to black (0 tone).

The image is changed after use of “Binary” filter.

3.13.6. Adjustment for Detection Conditions

3.13.6.1. Detection Color

Detection Conditions

Detection Color

☐ White

☒ Black

Selects which area ([Black] or [White]) of the binarized image to measure.

(Reference) When "Color to Binary" mode is selected, the extracted color becomes [White].

[Detection Color]

(General Overview)

"Detection Color" specifies the color (either black or white) whose area will be measured in a binary image.



(Usage)

When the area to be measured is white, set this parameter to "White", and for black, set it to "Black".

Detection Color

☒ White

☐ Black

The area of the white part of the mold is found.

	Measured
Area	1097

Detection Color

☐ White

☒ Black

The area of the black part of the mold is found.

	Measured
Area	6803

3.13.7. Adjustment for Judgment Conditions

Judgment Conditions	
◦ Area	
Upper Limit	-----
Measured	19392
Lower Limit	00015000

Set the range (upper/lower limit) judged as OK for the measured value.

The item(s) of judgment conditions set in this tool is/are as follows:

Area

The [Measured] field shows the measured result for the current image.

It is a useful reference for setting the upper/lower limit.

3.13.7.1. Defaults List of Judgment Conditions

Judgment Conditions Name	Defaults		On Change		Memo
	Upper Limit	Lower Limit	Upper Limit	Lower Limit	
Area	-----	15000			

3.14. (CAM2) [T206]SCREW-1



3.14.1. Summary

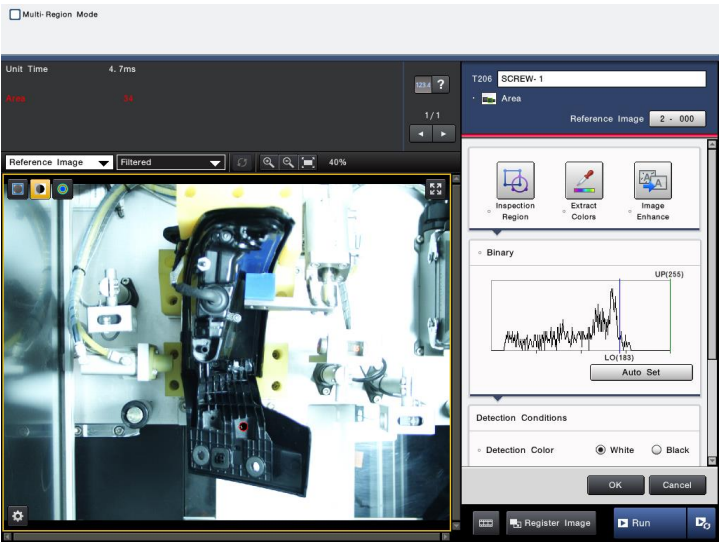
Summary of this tool
Measures the "White" or "Black" area after binarization (black-and-white processing) of the captured image.
Memo

3.14.2. Defaults List of Parameters

Parameter Name	Defaults	On Change	Memo
Upper Binary Threshold	255		
Lower Binary Threshold	183		
Detection Color	White		

3.14.3. Tool Edit Screen View

This is the edit screen for the [T206]SCREW-1 tool.



[Inspection Region]

Sets the inspection region. For details, refer to [3.14.4 Setting the Inspection Region](#).



[Extract Colors]

Sets the color extraction settings. [Gray] is used for the extract color mode in this tool.



[Image Enhance]

Sets the image enhancement settings. The type(s) of Image Enhance used in this tool

is/are as follows. Refer to [4.2 Image Enhance Settings](#) for the summary of each Image Enhance type.

Image Enhance List

0 Binary

[Binary]

Sets the binary settings. For details, refer to [3.14.5 Adjustment for Binary](#).

[Detection Conditions]

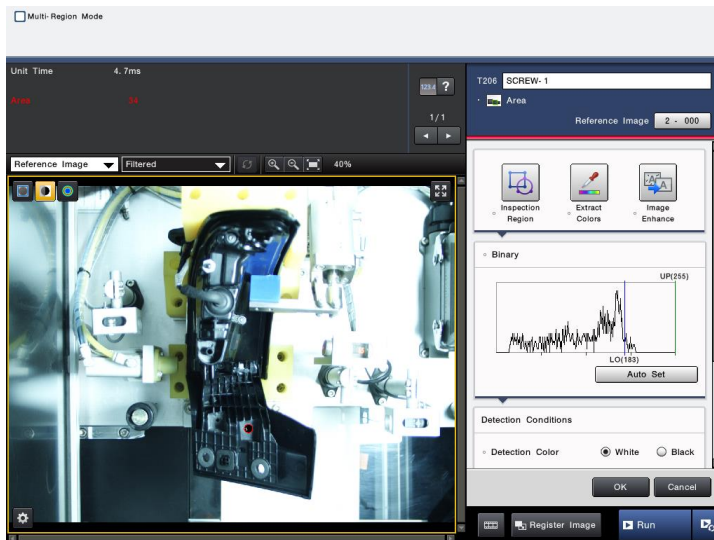
Sets the detection conditions according to the measurement items. For details, refer to [3.14.6 Adjustment for Detection Conditions](#).

[Judgment Conditions]

Sets the judgment conditions according to the measurement item. For details, refer to [3.14.7 Adjustment for Judgment Conditions](#).

3.14.4. Setting the Inspection Region

3.14.4.1. Inspection Region



Specifies the region to be measured.

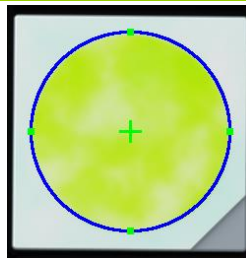
Circle is used for specifying the inspection region in this particular case.

[Circle]

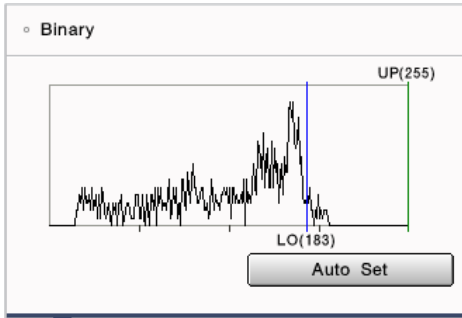
(How to setup)

Specify three points on the target circle.

Drag the green points to change the size, and drag from the inside of the circle to move the location.



3.14.5. Adjustment for Binary

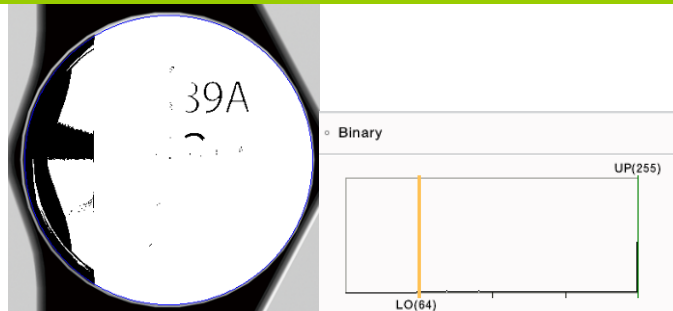


Specifies the threshold value of the upper (UP) and lower (LO) limit on the histogram showing the concentration distribution in the inspection region to binarize the image as follows: The pixels having a intensity within the range of the threshold are converted to white and other pixels to black.

[Binary]

(General Overview)

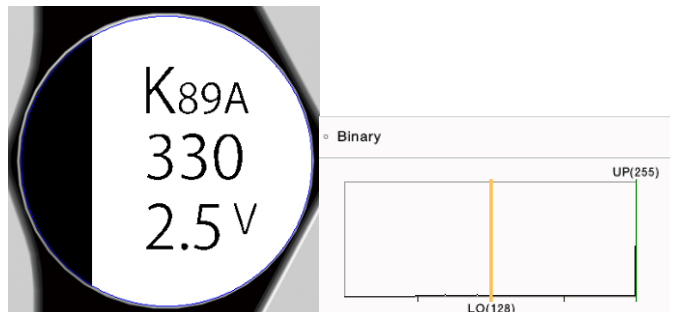
“Binary” is a filter which binarizes an image to white (255 tones) and black (0 tone). This filter is mainly used in “Area” and “Blob” tools.



(Usage)

↓ Change the LO value from 64 to 128.

Basically, change the “LO” value to set the range where brightness is converted to white (255 tones) and the range where brightness is converted to black (0 tone).



“LO/UP”: Sets brightness falling in the range between “LO” and “UP” to white (255 tones), and brightness out of the range to black (0 tone).

The image is changed after use of “Binary” filter.

3.14.6. Adjustment for Detection Conditions

3.14.6.1. Detection Color

Detection Conditions	
◦ Detection Color	☒ White ☐ Black

Selects which area ([Black] or [White]) of the binarized image to measure.

(Reference) When "Color to Binary" mode is selected, the extracted color becomes [White].

[Detection Color]

(General Overview)

"Detection Color" specifies the color (either black or white) whose area will be measured in a binary image.



(Usage)

When the area to be measured is white, set this parameter to "White", and for black, set it to "Black".

Detection Color	
☒ White	☐ Black

The area of the white part of the mold is found.

Measured	
Area	1097

Detection Color	
☐ White	☒ Black

The area of the black part of the mold is found.

Measured	
Area	6803

3.14.7. Adjustment for Judgment Conditions

Judgment Conditions	
◦ Area	
Upper Limit	00000750
Measured	34
Lower Limit	00000104

Set the range (upper/lower limit) judged as OK for the measured value.

The item(s) of judgment conditions set in this tool is/are as follows:

Area

The [Measured] field shows the measured result for the current image.

It is a useful reference for setting the upper/lower limit.

3.14.7.1. Defaults List of Judgment Conditions

Judgment Conditions Name	Defaults		On Change		Memo
	Upper Limit	Lower Limit	Upper Limit	Lower Limit	
Area	750	104			

3.15. (CAM2) [T207]SCREW-2



3.15.1. Summary

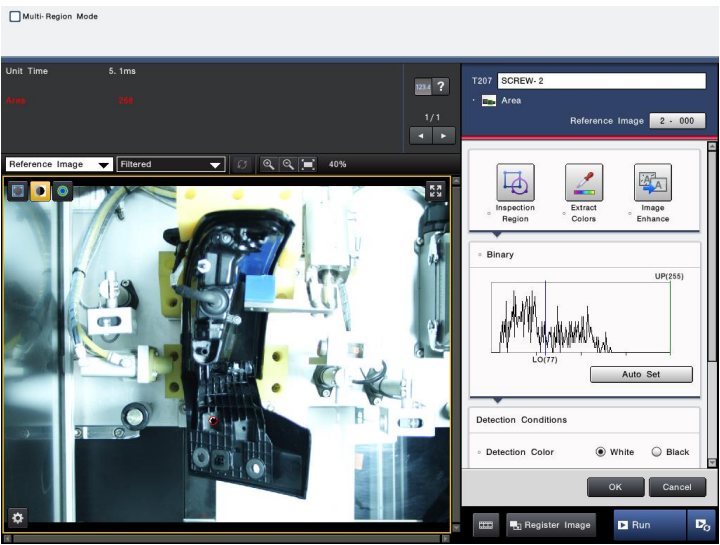
Summary of this tool
Measures the "White" or "Black" area after binarization (black-and-white processing) of the captured image.
Memo

3.15.2. Defaults List of Parameters

Parameter Name	Defaults	On Change	Memo
Upper Binary Threshold	255		
Lower Binary Threshold	77		
Detection Color	White		

3.15.3. Tool Edit Screen View

This is the edit screen for the [T207]SCREW-2 tool.



[Inspection Region]

Sets the inspection region. For details, refer to [3.15.4 Setting the Inspection Region](#).



[Extract Colors]

Sets the color extraction settings. [Gray] is used for the extract color mode in this tool.



[Image Enhance]

Sets the image enhancement settings. The type(s) of Image Enhance used in this tool

is/are as follows. Refer to [4.2 Image Enhance Settings](#) for the summary of each Image Enhance type.

Image Enhance List
0 Binary

[Binary]

Sets the binary settings. For details, refer to [3.15.5 Adjustment for Binary](#).

[Detection Conditions]

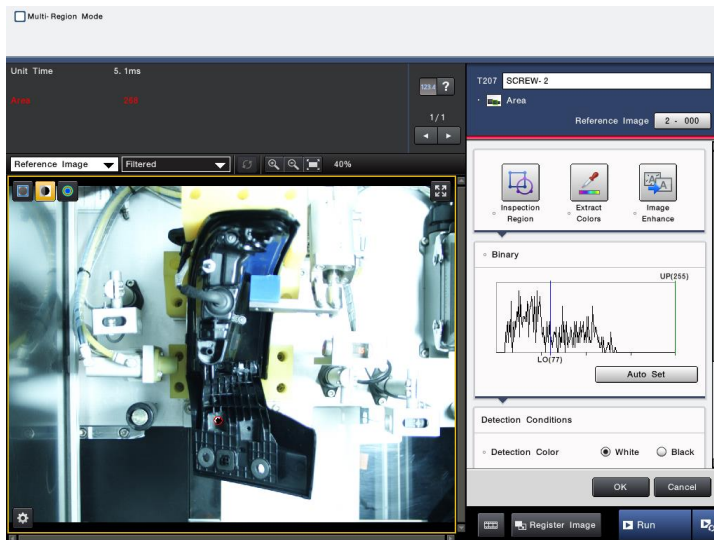
Sets the detection conditions according to the measurement items. For details, refer to [3.15.6 Adjustment for Detection Conditions](#).

[Judgment Conditions]

Sets the judgment conditions according to the measurement item. For details, refer to [3.15.7 Adjustment for Judgment Conditions](#).

3.15.4. Setting the Inspection Region

3.15.4.1. Inspection Region



Specifies the region to be measured.

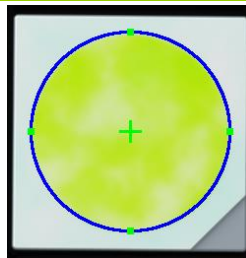
Circle is used for specifying the inspection region in this particular case.

[Circle]

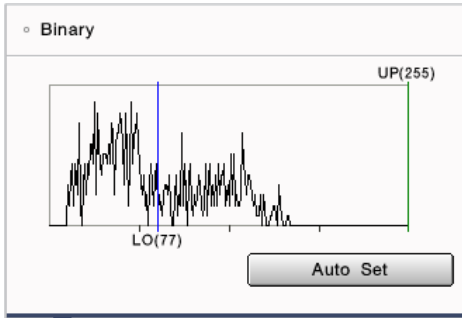
(How to setup)

Specify three points on the target circle.

Drag the green points to change the size, and drag from the inside of the circle to move the location.



3.15.5. Adjustment for Binary

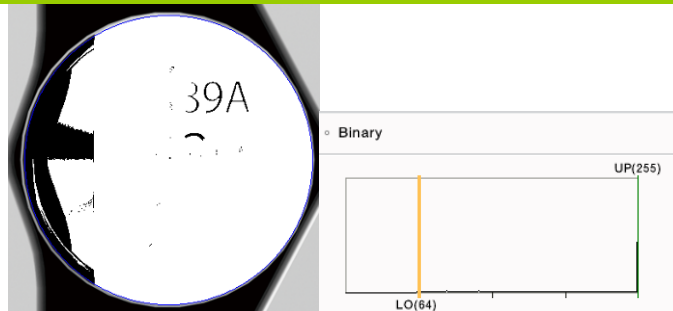


Specifies the threshold value of the upper (UP) and lower (LO) limit on the histogram showing the concentration distribution in the inspection region to binarize the image as follows: The pixels having a intensity within the range of the threshold are converted to white and other pixels to black.

[Binary]

(General Overview)

“Binary” is a filter which binarizes an image to white (255 tones) and black (0 tone). This filter is mainly used in “Area” and “Blob” tools.

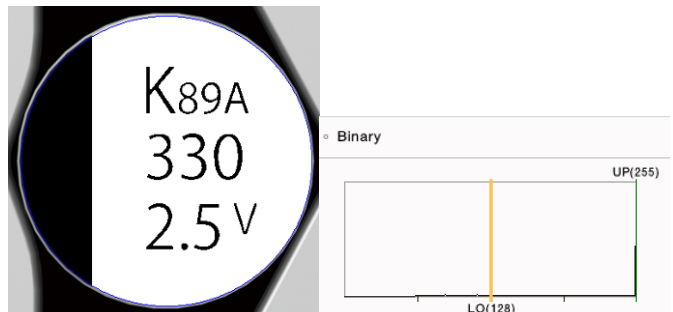


(Usage)

↓ Change the LO value from 64 to 128.

Basically, change the “LO” value to set the range where brightness is converted to white (255 tones) and the range where brightness is converted to black (0 tone).

“LO/UP”: Sets brightness falling in the range between “LO” and “UP” to white (255 tones), and brightness out of the range to black (0 tone).



The image is changed after use of “Binary” filter.

3.15.6. Adjustment for Detection Conditions

3.15.6.1. Detection Color

Detection Conditions

Detection Color

☒ White

☐ Black

Selects which area ([Black] or [White]) of the binarized image to measure.

(Reference) When "Color to Binary" mode is selected, the extracted color becomes [White].

[Detection Color]

(General Overview)

“Detection Color” specifies the color (either black or white) whose area will be measured in a binary image.



(Usage)

When the area to be measured is white, set this parameter to “White”, and for black, set it to “Black”.

Detection Color

☒ White

☐ Black

The area of the white part of the mold is found.

	Measured
Area	1097

Detection Color

☐ White

☒ Black

The area of the black part of the mold is found.

	Measured
Area	6803

3.15.7. Adjustment for Judgment Conditions

Judgment Conditions	
◦ Area	
Upper Limit	-----
Measured	268
Lower Limit	00000401

Set the range (upper/lower limit) judged as OK for the measured value.

The item(s) of judgment conditions set in this tool is/are as follows:

Area

The [Measured] field shows the measured result for the current image.

It is a useful reference for setting the upper/lower limit.

3.15.7.1. Defaults List of Judgment Conditions

Judgment Conditions Name	Defaults		On Change		Memo
	Upper Limit	Lower Limit	Upper Limit	Lower Limit	
Area	-----	401			

4. Supplement

4.1. Extract Colors Settings

[Color to Binary]

<General Overview>

“Color to Binary” is an extraction method used to convert a captured color image into a binary image by defining a selected color as white (255 tones) and all the others as black (0 tone).

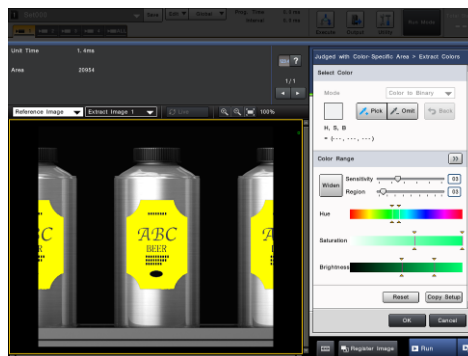
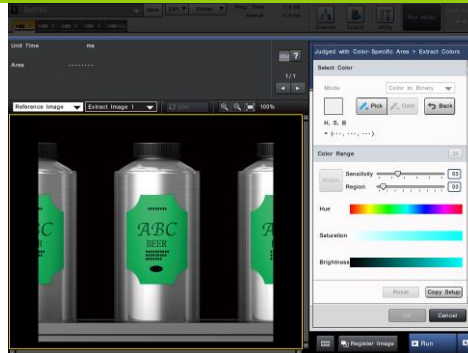
<Basic Procedure> (See the figure to the right.)

Click “Pick” and move the mouse cursor (Dropper) to a part bearing the color to be extracted on the image. Then, clicking it extracts the color and sets it to the selected color range. The display shows a through image where the selected color is in yellow and other colors are shown in gray.

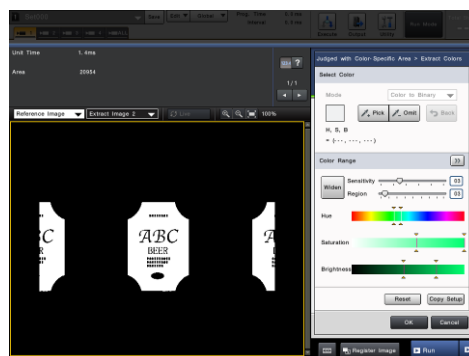
When the extraction is insufficient, change the dropper point and repeat the procedure.

<If the selected color range is excessively extended...>

“Omit” excludes clicked colors from the selected colors. “Back” can cancel the immediate previous “Pick”/“Omit”/“Widen” operation for one time. “Reset” sets back the selected color range to the default setting.



The selected color is indicated in yellow.

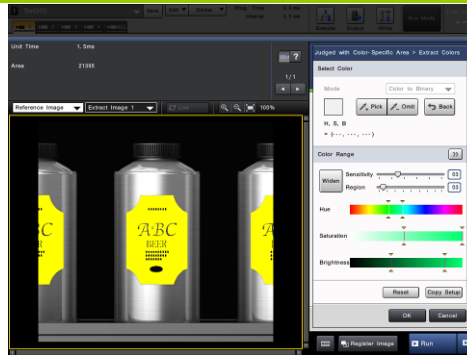


Change the display type to “Extract Image 2”, and the final image after the color extraction can be checked.

[Color to Binary]

<To automatically extend “Color Range” settings...> (See the figure to the right.)

Use “Widen” to extend the selected color range automatically.



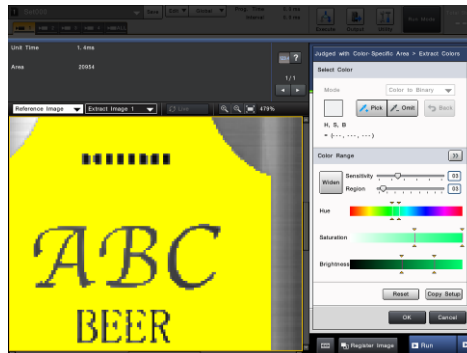
Press the “Widen” button to extend the selected color range to be extracted.

<To set in greater detail...> (See the figure to the right.)

“Sensitivity” specifies a size of “Color Range” every time “Pick”/“Omit”/“Widen” is pressed.

“Region” specifies a region size for selection every time “Pick”/“Omit” is pressed.

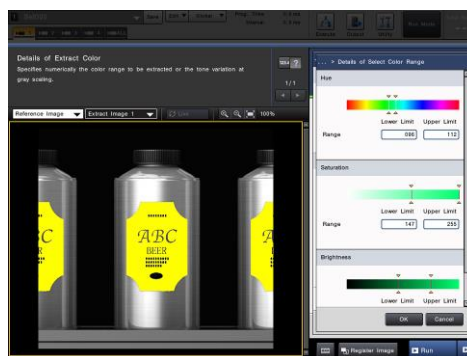
“Zoom” or the mouse wheel zooms-in/-out the display.



More detailed setting can be performed by zooming in the image.

[>>] specifies a desired color by a numerical value. The upper and lower limits can be set for three parameters of Hue (H), Saturation (S) and Brightness (B).

For example, after extending the B range, colors, whose H and S are the same but brightness is different, can be selected.



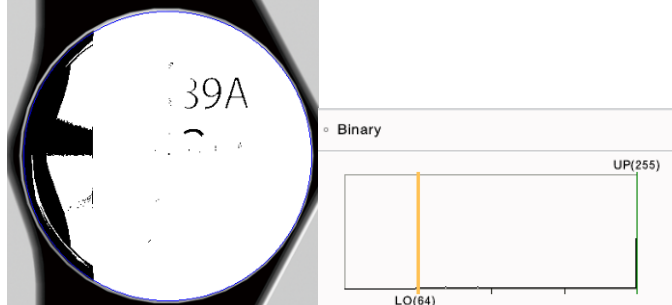
A color can be selected by specifying a numerical value.

4.2. Image Enhance Settings

[Binary]

(General Overview)

“Binary” is a filter which binarizes an image to white (255 tones) and black (0 tone). This filter is mainly used in “Area” and “Blob” tools.

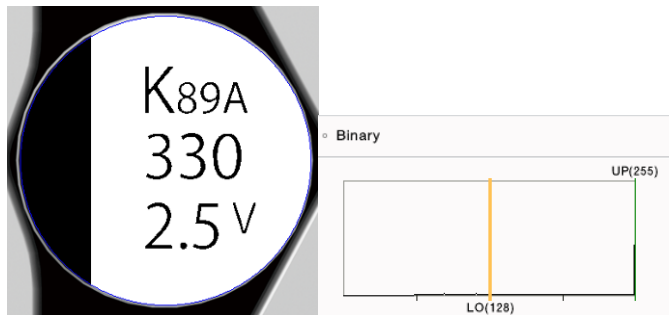


(Usage)

↓ Change the LO value from 64 to 128.

Basically, change the “LO” value to set the range where brightness is converted to white (255 tones) and the range where brightness is converted to black (0 tone).

“LO/UP”: Sets brightness falling in the range between “LO” and “UP” to white (255 tones), and brightness out of the range to black (0 tone).



The image is changed after use of “Binary” filter.

5. Appendix

5.1. Program Export/Import

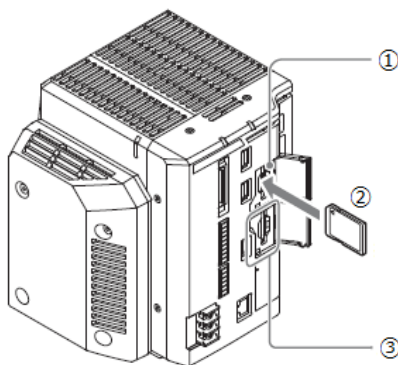
This appendix serves as a guide on exporting program setting files and image files to the SD Card.

By following the operation procedure in this appendix, files needed for simulation in remote locations are exported to the SD Card.

5.1.1. Preparation

Inserting an SD Card

Insert the SD card into the SD1 and SD2 slot such that the triangle inscription is on the top of the card.



1. SD2 slot
2. Insert SD card.
3. SD1 slot

(Do not insert or remove while power is on.)

Supported SD Card Models

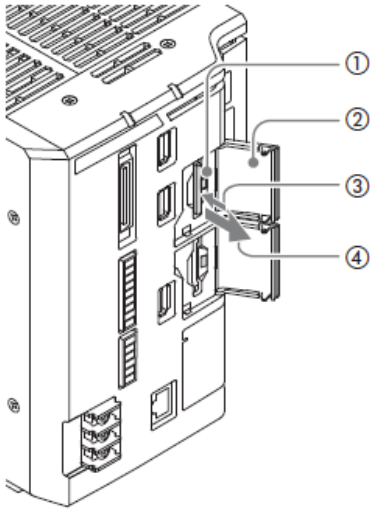
Use the Keyence optional OP-87133 (512 MB), optional CA-SD1G (1 GB), optional CA-SD4G (4 GB), or optional CA-SD16G (16 GB). Note that the controller ships with an OP-87133 (CA-SD1G for CV-X350/X450/X470/X480/X480LJ and CA-SD4G for CV-X490) inserted in the SD1 slot.

Note: SD cards other than the models mentioned above may not operate normally and are exempt from the warranty of this equipment.

Removing an SD Card



Choose [Utility] > [Remove External Media] > [Remove SD Card 2] to turn off the power of SD Card 2. Then, press the SD card in the SD2 slot inward to release and remove the card.



1. Make sure the access lamp is no longer lit.
2. Open cover.
3. Press the SD card to unlock it.
4. Remove SD card.

5.1.2. Archived Image

What is [Archived Image]

An archived image is an image temporarily saved in the internal memory of the CV-X. Images are archived and accumulated during operation.

When performing [Exporting the Current Program with Archived Images] and [Exporting Archived Images Only], it is necessary that there are images archived. Verify the images using the procedure below.

Accumulating Archived Images

Images are automatically archived whenever a trigger is input during Run Mode.

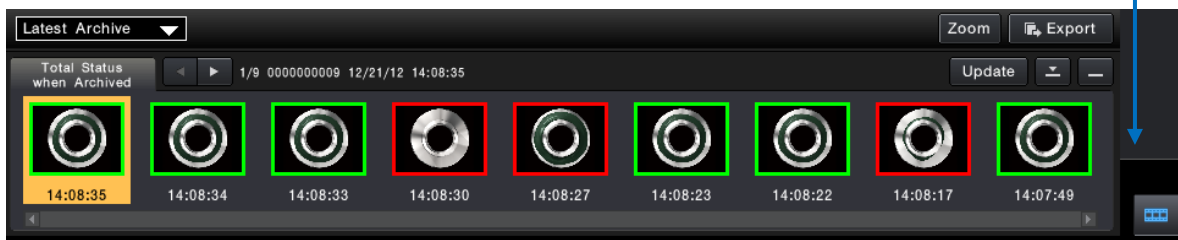


Click the button on the upper right portion of the screen to switch to Run Mode and input a trigger.

Verifying Archived Images

Click the button on the lower right portion of the screen to open the Image Strip.

If there were images archived, they will be displayed as thumbnails.



5.1.3. Export

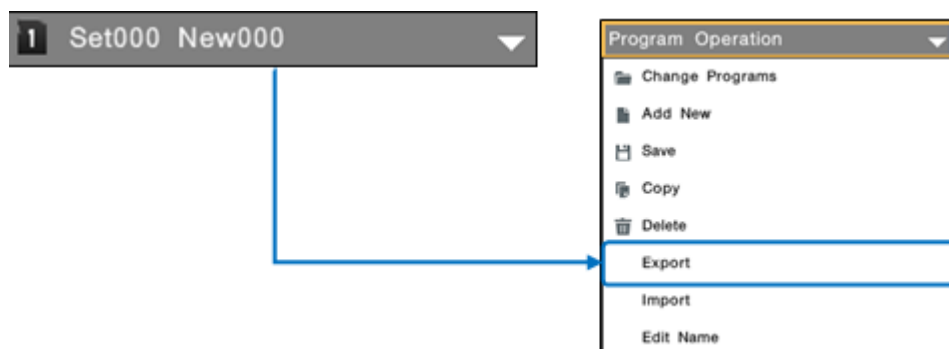
Exporting a Program Setting

This is the procedure for exporting a program setting to the SD card.



Click the button on the upper right portion of the screen to switch to Setup Mode.

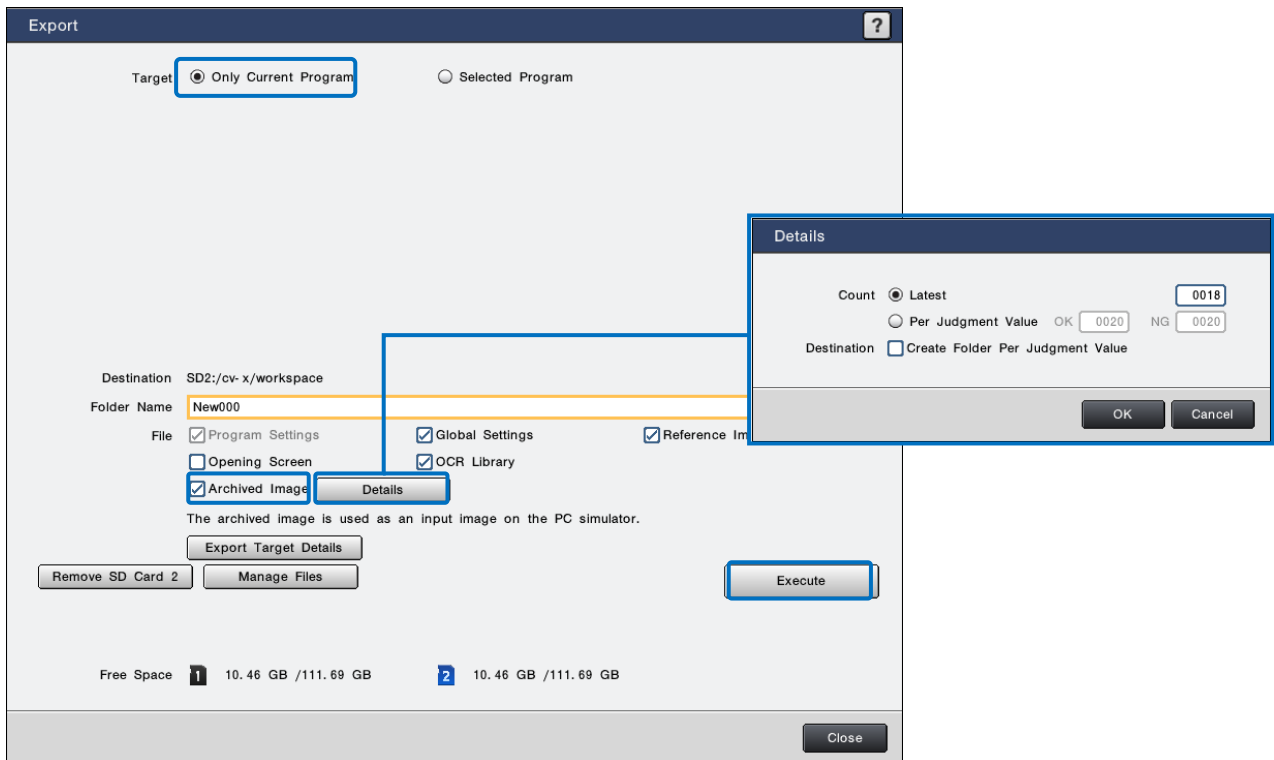
Click the drop-down menu on the upper left of the screen and choose [Export] from the Program Operation menu that appears.



(1) In the Case of Exporting the Current Program with Archived Images

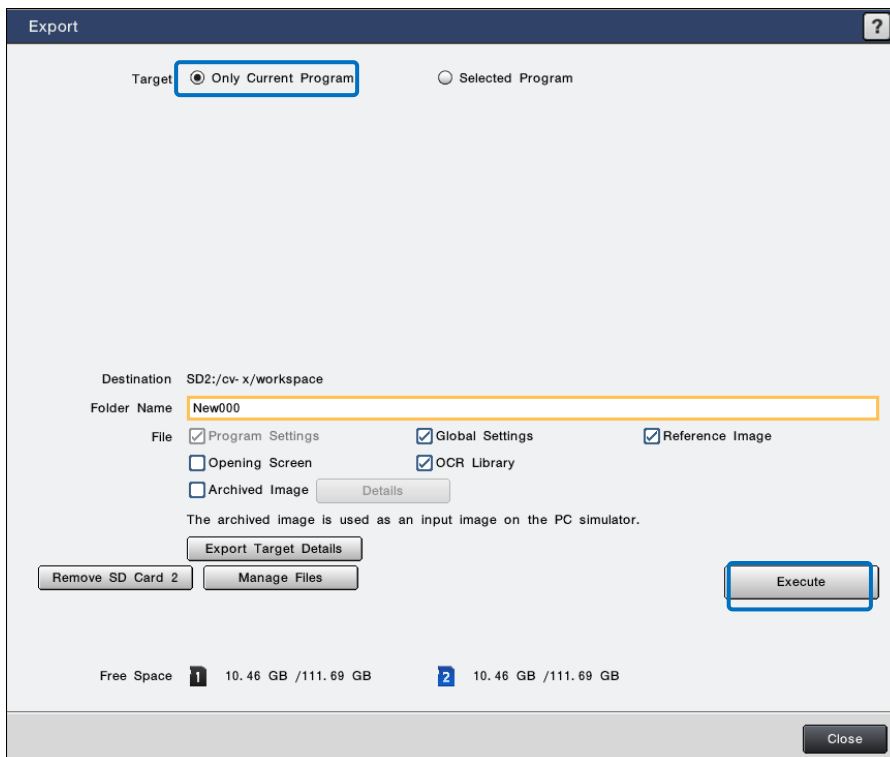
Select [Only Current Program]. Check [Archived Image] under [File], left-click , and specify the number of images to export. Then, click the [Execute] button.

Note: The latest archived images of the specified number are exported together with the program setting. Decide on the number of images to export while paying attention to the remaining capacity of the SD card.



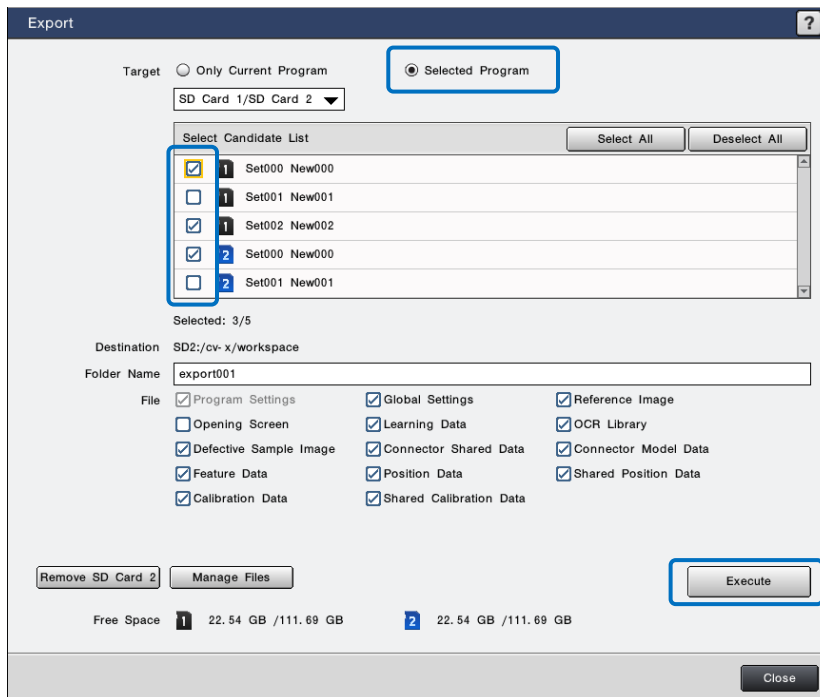
(2) In the Case of Exporting the Current Program Only

Select [Only Current Program] and click on the [Execute] button. Choose the target files to export as necessary.



(3) In the Case of Exporting Selected Programs

Select [Selected Program]. Put a check mark on the desired program settings to export and then click the [Execute] button. Choose the target files to export as necessary.



Details of Export Target [File]

File	Explanation
Program Settings	The program setting file (inspect.dat) of tools etc.
Global Settings	The global settings data (env.dat) such as Input/Output etc.
Reference Image	The reference images (ref*_*.bmp [.jpg] [.png]).
Opening Screen	The display image data (Logo.bmp) shown at start-up.
OCR Library	The library data (dic*.dat) used in the OCR tool.
Archived Image	Exports the specified number of latest or specified type of archived images (Available only when [Only Current Program] is selected as the Target).
Learning Data	The learning data (SCT.tbd) of Auto-Teach Inspection.
Defective Sample Image	The images (No_NGMASTER_CAM No_NG.bmp [.jpg] [.png]) set as defective samples in Auto-Teach Inspection.

Feature Data	The feature data (STF.tbd) set in Feature Drawing Tool.
Shape Region Data	The shape region data (RGN_SHP.dat) set in Shape Region.
Outline Setting Data	The outline setting data (CNT_SHP.dat) set in Outline Comparison.
Subtraction Setting Data	The subtraction setting data (TDS_MDL.dat) set in 3D Comparison.
Camera Calibration Data	The calibration data (TDC_G_CLB_*.dat) used in 3D Capture Mode.
Position Data	The position data (RBT_L_POS_*.dat) used in Vision-Guided Robotics Setting.
Shared Position Data	The shared position data (RBT_G_POS_*.dat) used in Vision-Guided Robotics Setting.
Calibration Data	The calibration data (RBT_L_CLB_*.dat, 3D_RBT_L_CLB_*.dat) used in Vision-Guided Robotics Setting.
Shared Calibration Data	The shared calibration data (RBT_G_CLB_*.dat, 3D_RBT_G_CLB_*.dat) used in Vision-Guided Robotics Setting.
Layout Data	The layout data (LYT_G_*.tbd) used in 3D Vision-Guided Robotics Setting.
Robot Model Data	The robot model data (RBT_G_RMD_*.dat) used in 3D Vision-Guided Robotics Setting.
Connector Model Data	The model data (mdl_l_*.tbd) registered in the Connector Dimension Inspection tool.
Connector Shared Data	The data (mdl_s_*.tbd) registered in the Share Settings of the Connector Dimension Inspection tool.

Exporting Archived Images Only

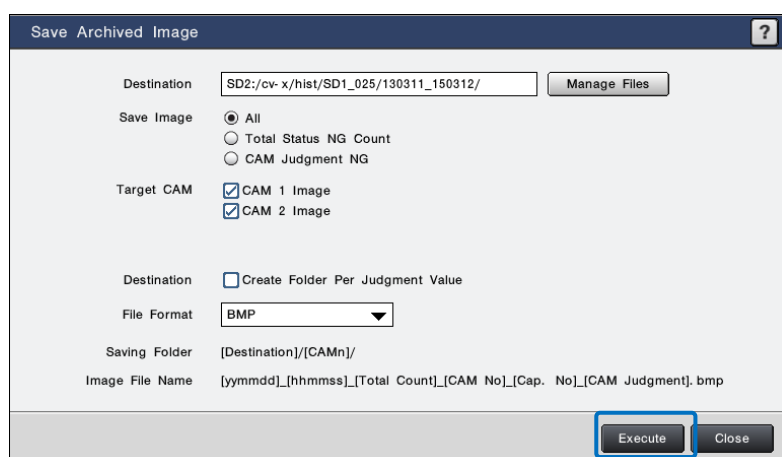
Click the button  on the lower right of the screen to open the Image Strip. Then, click on the [Export] button to make the images selectable.



Check the desired images for export (Pressing the [Select All] button puts a check mark on all the images). Upon clicking the [Execute] button, the [Save Archived Image] dialog appears.



Make the appropriate selection for [Save Image], [Target CAM], [Destination] and [File Format] and then click on the [Execute] button.



Save Image (This can be set when [Select All] is chosen in the Image Strip)

Option	Explanation
All	Exports all the images.
Total Status NG Count	Exports the images of all cameras of measurement counts where the Total Status was NG during archiving.
CAM Judgment NG	Exports only the images where CAM Judgment was NG during archiving.

5.1.4. Import

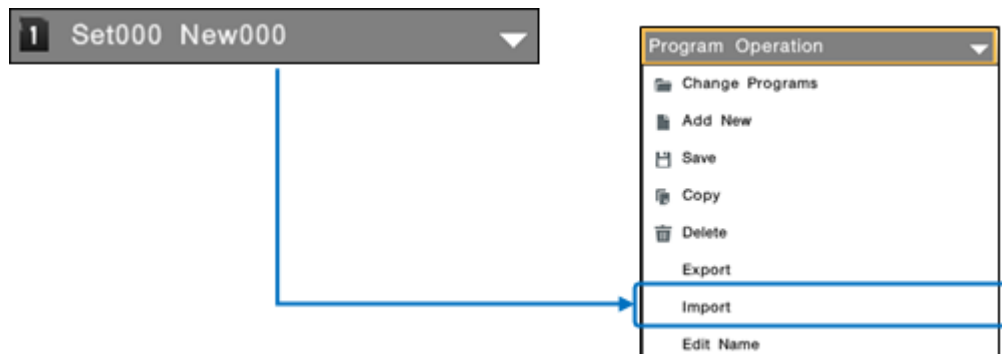
Importing a Program Setting

This is the procedure for importing a program setting from the SD card.



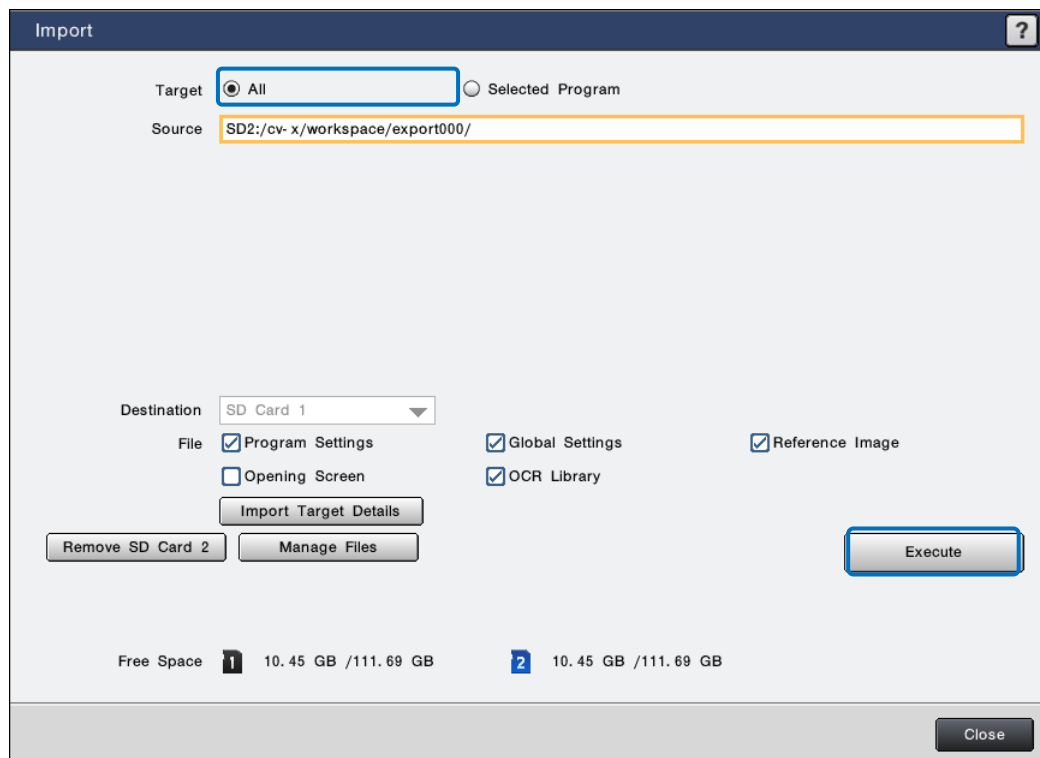
Click the button on the upper right portion of the screen to switch to Setup Mode.

Click the button on the upper left of the screen and choose [Import] from the Program Operation menu that appears.



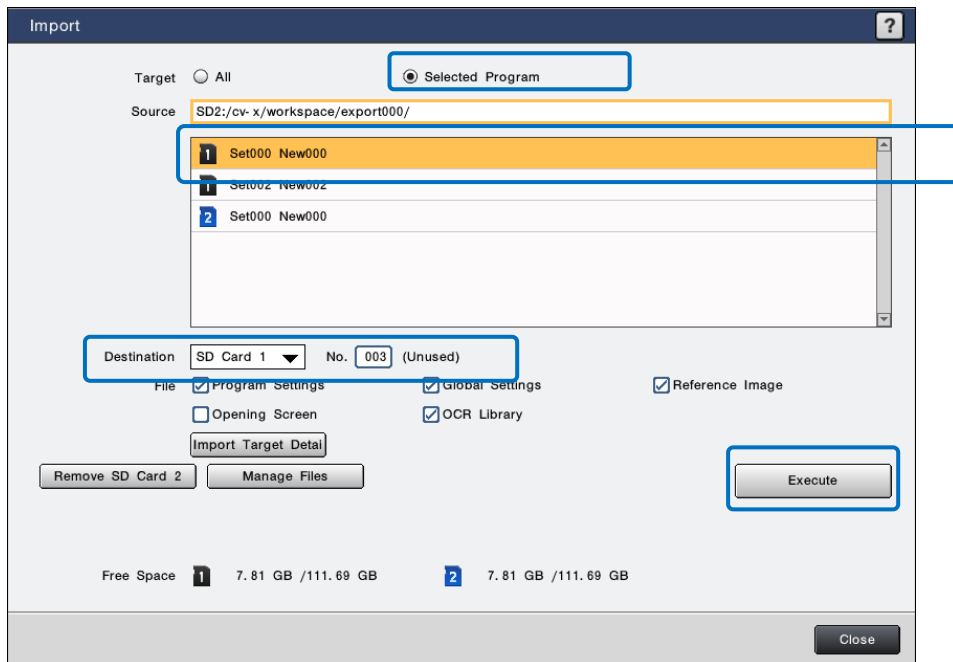
(1) In the Case of Importing All Programs Within the Workspace

Select [All]. Specify the target workspace folder in [Source] and then click the [Execute] button. Choose the target files to import as necessary.



(2) In the Case of Importing a Selected Program from the Workspace

Select [Selected Program] and specify the target workspace folder in [Source]. Choose the desired program setting to import from the list, specify the [Destination] to put it to, and then click the [Execute] button. Choose the target files to import as necessary.



Details of Import Target [File]

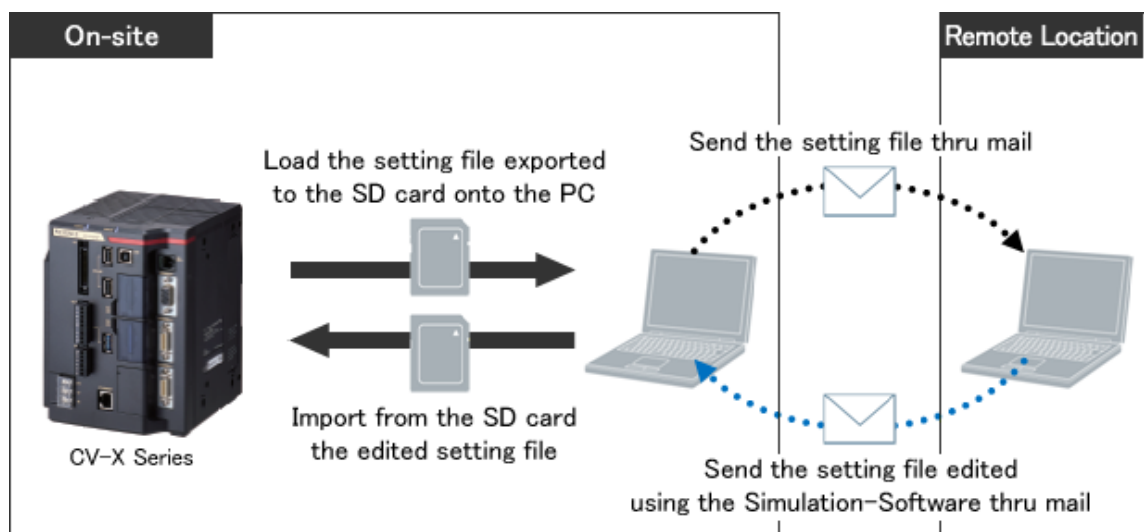
File	Explanation
Program Settings	The program setting file (inspect.dat) of tools etc.
Global Settings	The global settings data (env.dat) such as Input/Output etc.
Reference Image	The reference images (ref*_*.bmp [.jpg] [.png]).
Opening Screen	The display image data (Logo.bmp) shown at start-up.
OCR Library	The library data (dic*.dat) used in the OCR tool.
Learning Data	The learning data (SCT.tbd) of Auto-Teach Inspection.
Defective Sample Image	The images (No_NGMASTER_CAM No_NG.bmp [.jpg] [.png]) set as defective samples in Auto-Teach Inspection.
Feature Data	The feature data (STF.tbd) set in Feature Drawing Tool.

Shape Region Data	The shape region data (RGN_SHP.dat) set in Shape Region.
Outline Setting Data	The outline setting data (CNT_SHP.dat) set in Outline Comparison.
Subtraction Setting Data	The subtraction setting data (TDS_MDL.dat) set in 3D Comparison.
Camera Calibration Data	The calibration data (TDC_G_CLB_*.dat) used in 3D Capture Mode.
Position Data	The position data (RBT_L_POS_*.dat) used in Vision-Guided Robotics Setting.
Shared Position Data	The shared position data (RBT_G_POS_*.dat) used in Vision-Guided Robotics Setting.
Calibration Data	The calibration data (RBT_L_CLB_*.dat, 3D_RBT_L_CLB_*.dat) used in Vision-Guided Robotics Setting.
Shared Calibration Data	The shared calibration data (RBT_G_CLB_*.dat, 3D_RBT_G_CLB_*.dat) used in Vision-Guided Robotics Setting.
Layout Data	The layout data (LYT_G_*.tbd) used in 3D Vision-Guided Robotics Setting.
Robot Model Data	The robot model data (RBT_G_RMD_*.dat) used in 3D Vision-Guided Robotics Setting.
Connector Model Data	The model data (mdl_l_*.tbd) registered in the Connector Dimension Inspection tool.
Connector Shared Data	The data (mdl_s_*.tbd) registered in the Share Settings of the Connector Dimension Inspection tool.

5.1.5. Exchange of Setting File

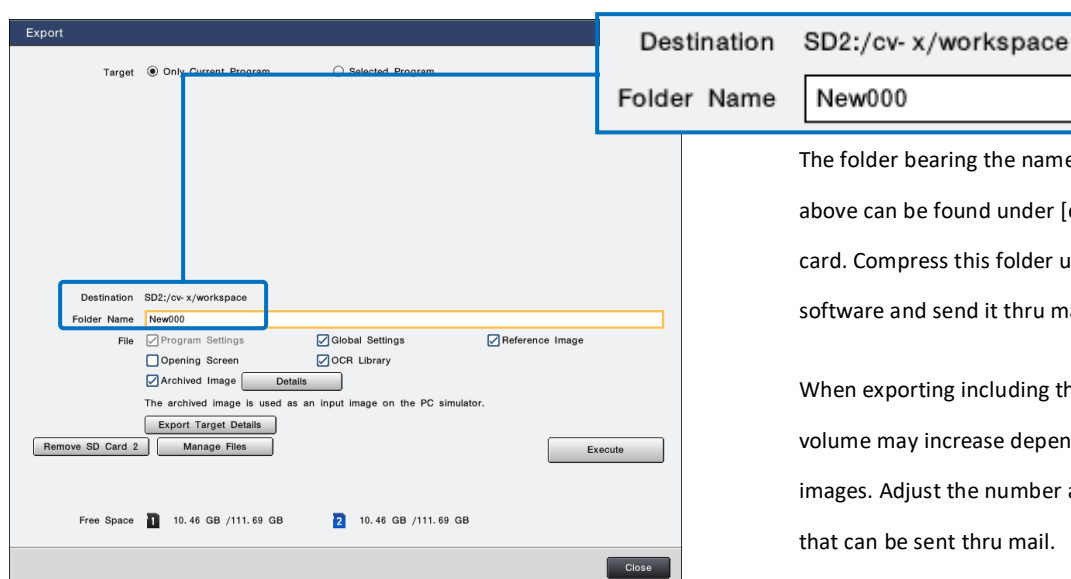
Sending the Setting File thru Mail

The exported setting file can be loaded into the [CV-X Series Simulation-Software] and can be edited and saved just as in a controller. For example, when product type addition or change of settings need to be made and the creator of the setting is in a remote location, the problem can be solved in the shortest possible time by exchange of the setting file thru mail.



Where the File (Exported Setting File) to Send thru Mail Is

The exported setting file is saved in the SD card in a folder structure called [Workspace] which can be loaded onto the [CV-X Series Simulation-Software] as is. When exchanging the setting file thru mail, it is convenient to send the [Workspace]. For example, when exporting a setting using the procedure described in [(1) In the Case of Exporting the Current Program with Archived Images] under section [3. Export]



The folder bearing the name specified in [Folder Name] above can be found under [cv-x]>[workspace] of the SD card. Compress this folder using a compression software and send it thru mail.

When exporting including the archived images, the volume may increase depending on the number of images. Adjust the number according to the amount that can be sent thru mail.

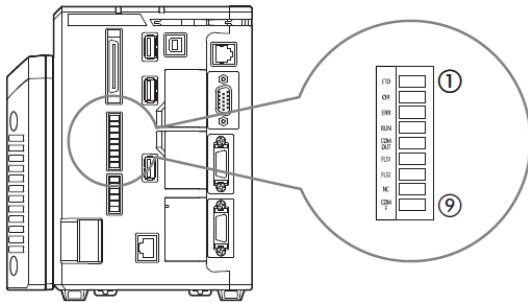
5.2. Outputting Total Status

5.2.1. Performing Wiring

When outputting the Total Status of the inspection, perform the wiring below.

Here, the wiring of the terminal blocks [OUT22: STO] and [OUT23: OR] to a PLC or a relay will be explained.

Controller Terminal Block (OUT Connector)



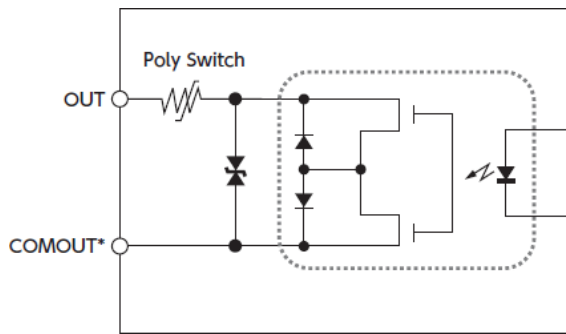
No.	Terminal Name *1
1	OUT22 (STO)
2	OUT23 (OR)
3	F_OUT2 (ERR)
4	F_OUT3 (RUN)
5	COMOUT1 (COMOUT)
6	F_OUT0 (FLS1)
7	F_OUT1 (FLS2)
8	N.C.
9	COMOUT_F (COM1)

Suitable wiring: AWG16 to 28

Terminal block screw torque: 0.25Nm or less

*1 The words in () are the words printed on the accompanying terminal blocks during shipment.

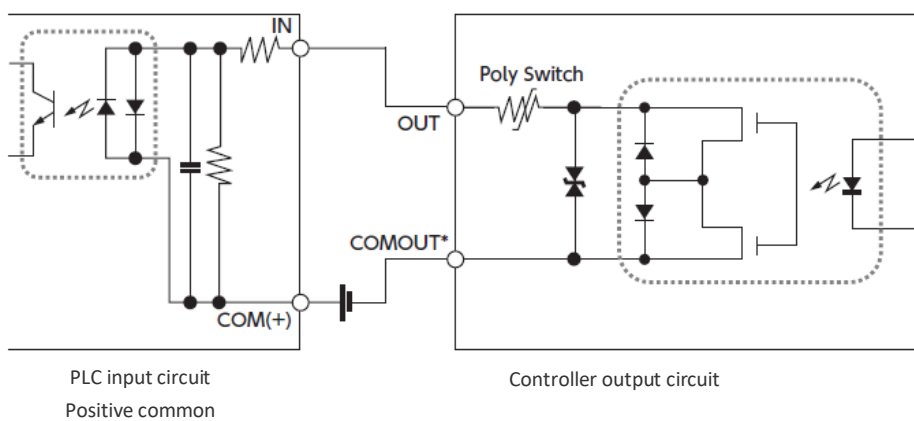
Output Circuit Diagram



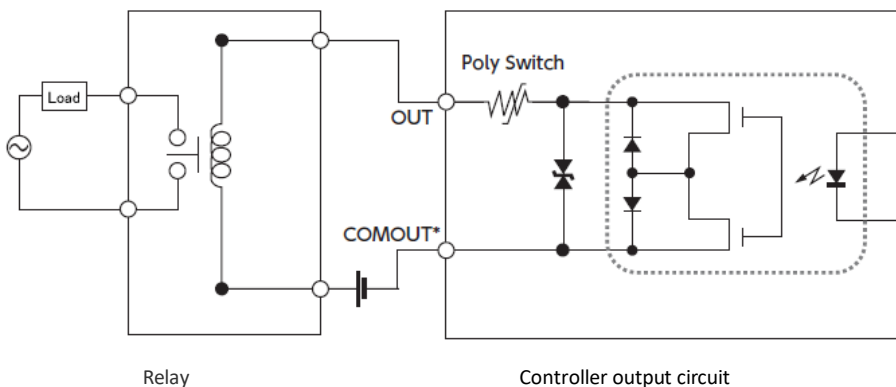
- Maximum applied voltage: 30V
- Maximum sink current: 50mA
- Leakage current: 0.1mA or less
- Residual voltage:
 - 1.4V or less (50mA)
 - 1.0V or less (20mA)

Examples of Connections

When connecting an output from the controller to a PLC with a positive common



When connecting an NPN output from the controller to a relay



5.2.2. Confirming External Terminal Connection Status (I/O Monitor)

The connection status of signals inputted/outputted via the CV-X400 /X300Series External Terminal can be confirmed.

Click the button on the upper right-hand side of the screen to switch to Setup Mode.

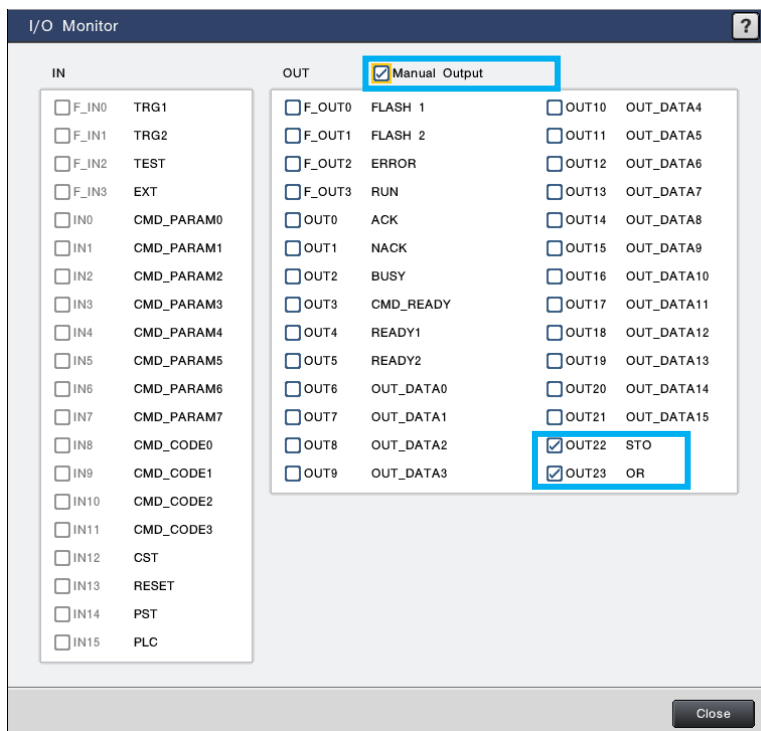


Select [I/O Monitor] from [Utility] to open the I/O Monitor.



The desired output terminals can forcibly be turned on and signal confirmation can be done on the connected device by putting a checkmark on [Manual Output]. Forced output can only be used in Setup Mode.

Put a checkmark on [OUT22 (STO)] and [OUT23 (OR)], and confirm whether the ON/OFF statuses of the signals can be recognized by the connected device (PLC, relay, etc.). Putting a checkmark turns the status to ON (Short) and removing the checkmark turns the status to OFF (Open).



When the ON/OFF status linked to checking/unchecking cannot be confirmed, check the wiring once again.

5.2.3. Setting the Tools to Include in Total Status Judgment

The tools to include in Total Status judgment can be arbitrarily selected. This is useful when performing of Total Status judgment using only the judgment results of particular tools is desired.

Click the button on the upper right-hand side of the screen to switch to Setup Mode.

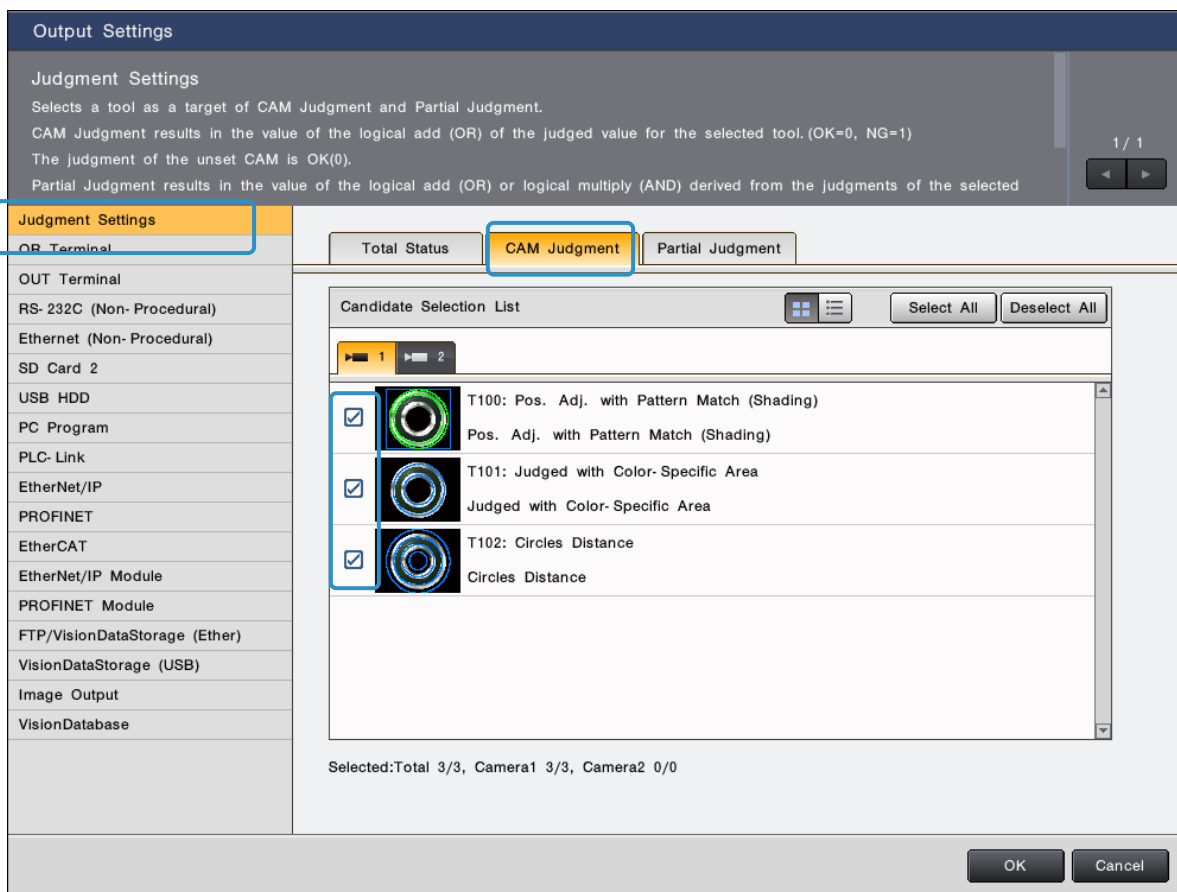


Open [Output].



Select [Judgment Settings] > [CAM Judgment].

In [Select Candidate List], verify that the tools to include in Total Status judgment are checked. (When using multiple cameras, switch between the camera tabs and perform the same verification.)



5.2.4. Setting OR Terminal Output Mode

When outputting the Total Status via the OR Terminal, select the terminal output mode.

Select [OR Terminal], then set the [OR Terminal Output Mode] and [OR Terminal Output Timing].

Output Settings

OR Terminal Output Settings

Selects the terminal output state when the total status value is output from the OR terminal.

- Update Every Judge (default): The terminal output state of the NG status varies in response to the judgment result at every final status.
- Hold Status: The NG status is maintained regardless of the judgment result after the NG status is output.

Judgment Settings

- OR Terminal**
- OUT Terminal
- RS-232C (Non- Procedural)
- Ethernet (Non- Procedural)
- SD Card 2
- USB HDD
- PC Program
- PLC- Link
- EtherNet/IP
- PROFINET
- EtherCAT
- EtherNet/IP Module
- PROFINET Module
- FTP/VisionDataStorage (Ether)
- VisionDataStorage (USB)
- Image Output
- VisionDatabase

OR Terminal Output Mode

- ☒ Update Every Judge
- ☐ Hold Status
- ☐ One Shot Output

One Shot Time (ms)

OR Terminal Output Timing

- ☒ Output on NG Status
- ☐ Output on OK Status

OK **Cancel**

OR Terminal Output Mode

- **Update Every Judge:** The OR terminal output state varies in response to the judgment result at every judgment finalization. Reading is synchronized to the rise of the STO outputted at every judgment finalization.

- **Hold Status:** Once an NG status is output, the NG status is maintained regardless of the judgment results that come after.

- **One Shot Output:** The OR terminal turns on only for the period set in [One Shot Time] and then it turns off after the elapse of the set time.

OR Terminal Output Timing

- **Output on NG Status**

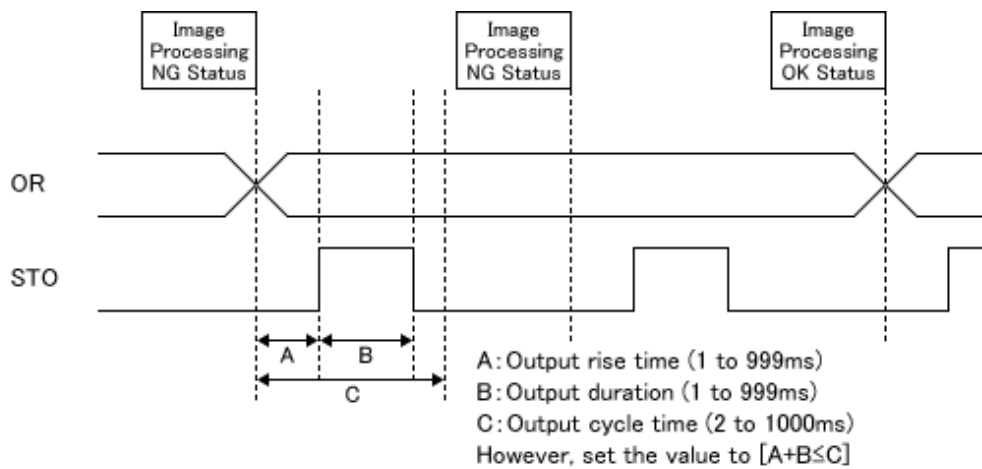
- **Output on OK Status**

Refer to the next page and following for the timing chart of each output mode.

5.2.5. Timing Chart

The timing chart for each OR terminal output mode can be seen below.

· Update Every Judge



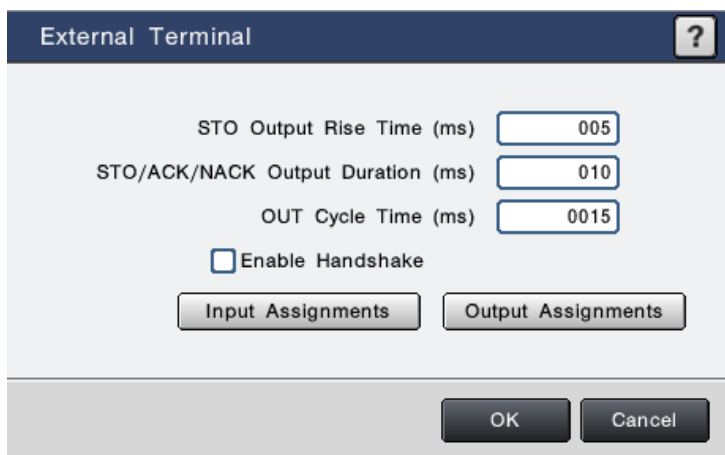
Since STO (OUT22) rises whenever a judgment is finalized, check the ON/OFF status of OR (OUT23) in synchronization to it. The STO output time (A/B/C in the timing chart) can be changed in the global settings. Adjust the time to a time that can be recognized by the connected PLC.

How to Change STO Output Time

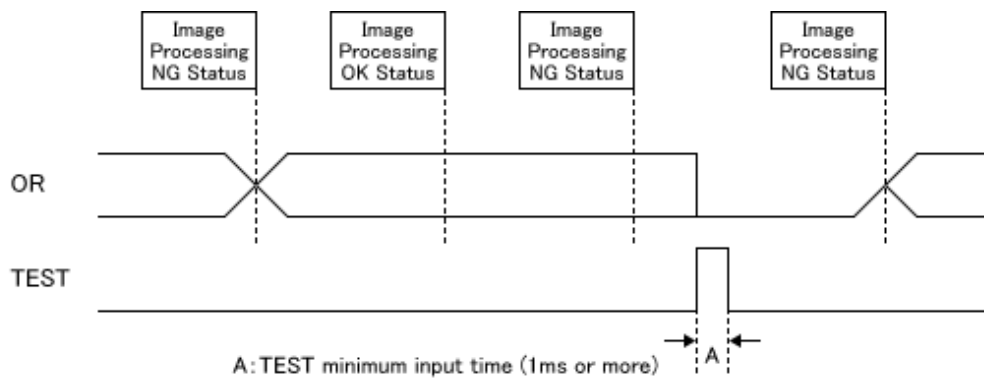
Select [Global] > [Communications & I/O] > [External Terminal].



Adjust A: Output rise time (1 to 999ms), B: Output duration (1 to 999ms) and C: Output cycle time (2 to 1000ms) to a time that can be recognized by the connected PLC. However, set the value to $A+B \leq C$.



· Hold Status

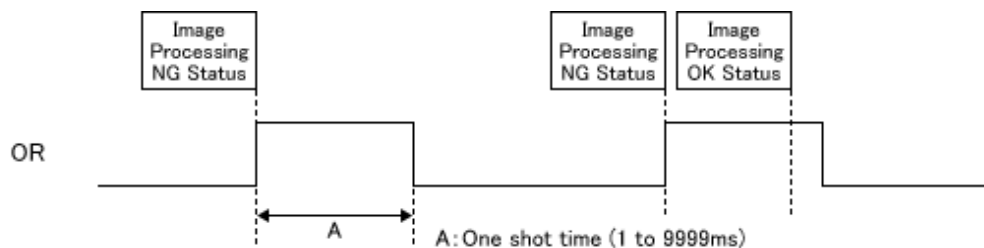


In normal circumstances, judgment output status is updated upon every trigger. But in this mode, after an NG Status occurs, it is held until it is released by TEST input etc.

Once an NG occurs, the NG status of the OR output does not change regardless of the actual judgment result.

When the TEST input is turned on/off, the hold status is released and all terminals are reset to the normal status. The OR output status will be updated when the result for the next OR output is output.

· One Shot Output



The OR output holds the NG status for the duration of the set time, and then releases the NG status afterwards. When continuous image processing is performed in a state where One Shot Output is set, there is a chance that the previous NG status is still held during the judgment result finalization time of STO-based image processing.

STD FENDER BOP CHECKING

Generated 2022-06-03